Int J Fract
https://doi.org/10.1007/s10704-019-00395-5

ORIGINAL PAPER

# A Coulomb-based model to simulate concrete cracking using cohesive elements

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Received: 10 February 2019 / Accepted: 19 October 2019
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**Abstract** This paper presents a constitutive model for cohesive elements to simulate crack formation and propagation in mode I and mixed mode in quasi-brittle materials. The model is based on the plasticity theory and features a Coulomb-based yield function, a non-associated flow rule and an implicit stress integration scheme. It also incorporates concepts of nonlinear fracture mechanics to model the softening behavior of the fracture process. Two softening laws are investigated, a bilinear and an exponential law. The results from concrete crack propagation simulations of several classical experiments are presented. Numerical load–CMOD and load–displacement curves showing softening behavior are in very good agreement when compared with experimental data. In summary, the model is able to provide accurate predictions of the whole fracture process on all cases studied.

**Keywords** Finite elements · Quasi-brittle materials · Concrete cracking · Cohesive elements · Plasticity

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## 1 Introduction

Modeling of quasi-brittle materials, such as concrete and rocks, proves to be a challenging task since it must take into account the effects of displacements and tension discontinuities caused by cracks. For this purpose, the literature includes several finite element techniques based on continuous and discontinuous approaches.

On the continuous approach, the cracks are represented in a smeared manner where the openings are distributed over the finite element. This class of models are frequently formulated considering orthogonal and non-orthogonal cracks; see e.g. (Gupta and Akbar 1984; de Borst and Nauta 1985; Barros and Figueiras 2001; Cunha et al. 2011, 2012; Dahlblom and Ottosen 1990; Červenka et al. 2018). Isotropic and anisotropic damage models are also examples of smeared crack models (Mazars 1986; Mazars and Pijaudier-Cabot 1996; Comi and Perego 2001; Basaran and Nie 2007; Pituba and Fernandes 2011). However, the continuous approach has limited ability to model crisp discontinuities, working better when cracks are scattered throughout the material as in some reinforced concrete (RC) applications. On the other hand, in the discontinuous approach, cracks are modeled as displacement discontinuities between finite elements; see e.g. (Ngo and Scordelis 1967; Nilson 1968; Bocca et al. 1990; Gerstle and Xie 1992; Li and Kaliakin 1993; Carol et al. 1997; Pandolfi et al. 1999; Cendón et al. 2000; Roy and Dodds 2001; Pandolfi and Ortiz 2002; Jin et al. 2003;

Published online: 12 November 2019

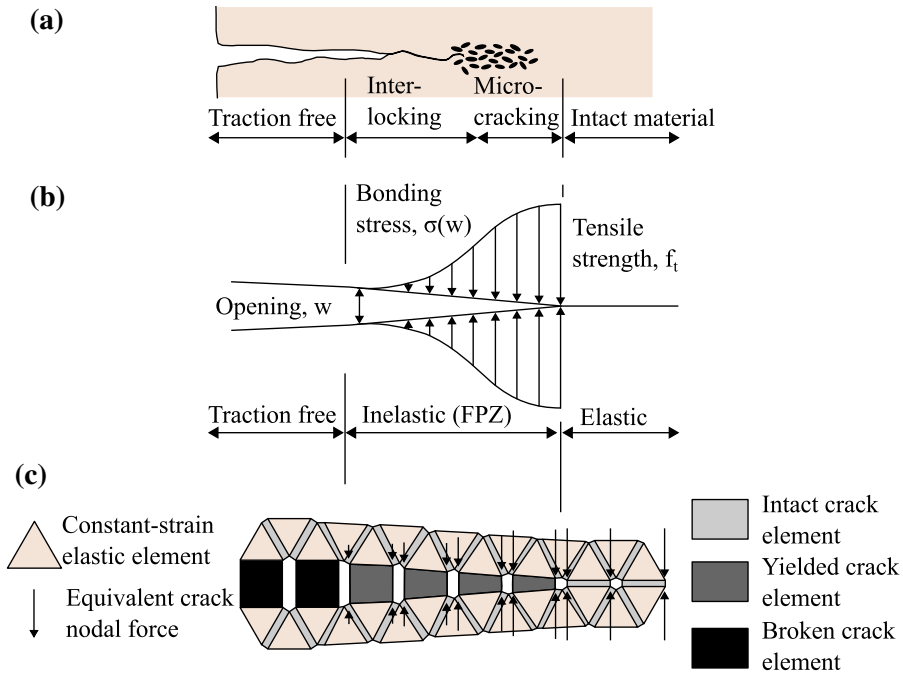



**Fig. 1** Concrete crack modeling using cohesive elements after Mahabadi et al. (2014): **a** conceptual model of a tensile crack in plain concrete; **b** theoretical fracture process zone (FPZ) model of Hillerborg et al. (1976); **c** discrete approach implementation of fracture process zone (FPZ) using triangular elastic elements and four-noded interface elements to represent the bulk material and the fracture, respectively

Song et al. 2006; Xie and Waas 2006; Roesler et al. 2007; Lens et al. 2009; Park et al. 2009; Segura and Carol 2010; Caggiano et al. 2012; Sánchez et al. 2014; Xu and Needleman 1994; Xu et al. 2015; Corona and Reedy 2011; Park and Paulino 2013; García-Álvarez et al. 2012; Morin et al. 2013; Xu et al. 2016; Tabiei and Zhang 2017). This approach usually requires the addition of zero-thickness interface elements intended to model the crack openings. This approach allows more realistic crack representations when compared with continuous models since it allows for the visualization of the crack openings due to the progressive detachment of bulk elements. The extended finite element method (X-FEM) (Melenk and Babuska 1996; Wells and Sluys 2001; Moës and Belytschko 2002; Mariani and Perego 2003; Asferg 2006; Fries and Belytschko 2006; Belytschko et al. 2009; Gupta et al. 2012; Simone 2007) is another technique that can be classified as a discrete approach since it tracks the crack paths during the analysis. By using finite element shape functions as partitions of unity (Wells and Sluys 2001), the displacement jump across a crack is represented by extra degrees of freedom at existing nodes. This allows for the convenient simulation of cracks evolutions without the need of remeshing. An additional discrete method that is worth mentioning is the one presented by (Manzoli et al. 2012, 2016) where thin interface elements with high aspect ratio are used as an alternative to cohesive elements. A simple constitutive damage model is used to describe the crack formation controlled by a prevalent mode I of fracture.

Regarding the use of cohesive elements in the modeling of plain concrete, the choice of the constitutive model is essential for an accurate representation of the fracture process. Most constitutive models use fracture mechanics concepts as the cohesive zone model (CZM) (Dugdale 1960; Barenblatt 1962; Hillerborg et al. 1976) to describe the relationship between stresses and crack widths as well as to identify the development of new cracks. The CZM includes a fracture process zone (FPZ) that has the ability to transfer stresses across the discontinuity (see Fig. 1). These stresses can be estimated by softening stress-opening laws. Also, the crack is assumed to propagate when the stress at the crack tip reaches the tensile strength.

This paper focuses on the development of a constitutive model for cohesive elements to be used in the simulation of quasi-brittle materials such as concrete. The model was elaborated using the plasticity theory featuring a Coulomb-based yield function, a non-associated flow rule, and an implicit stress integration scheme. Two nonlinear curves are used as stress-opening softening laws: a bi-linear law and the exponential Hordijk's

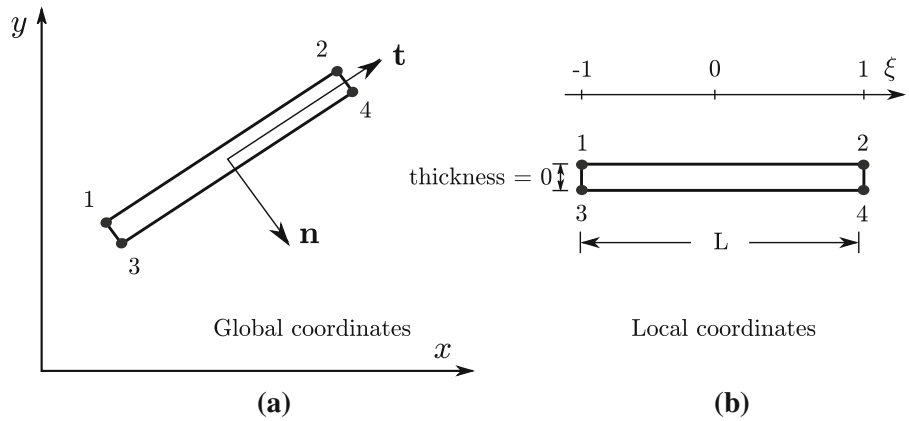



**Fig. 2** Four-node interface element: **a** global coordinate system $xy$ and local reference given by vectors **n** and **t**; **b** parametric coordinate system

law (Hordijk 1992). The constitutive model applied to cohesive elements allows simulation of the transfer of normal and shear stresses in a crack as well as the development of friction and slip between crack faces. These features allow the modeling of single and multi-cracks formation in mode I and mixed mode (I and II). Finite element simulations using the proposed model work for different levels of mesh refinement and elements with different interpolation orders. The model was validated in extension and shear tests and later applied in the simulation of several classical experiments of notched beams. These experiments were widely simulated by previous approaches using cohesive elements and other advanced methods as XFEM. However, several simulation results frequently present discrepancies predicting the softening branch of load–CMOD curves. Numerical results using the proposed model show very good agreement when compared with experimental results. In fact, peak and residual loads and CMOD values were very well predicted showing that it is possible to simulate the complete fracture process. This shows that the proposed model can be applied to a wide range of experiments. In addition, due to the use of an implicit integration scheme, the model provides numerical stability. No convergence problems were presented while running the simulations. All aspects of the model are carefully explored in order to allow the reproduction of the solution techniques and the development of finite element tools for large scale simulations in fields such as reinforced concrete and rock mechanics. The paper also presents a detailed procedure to introduce interface elements in a conventional mesh and the calculation of a representative element size needed in the constitutive model formulation.

## 2 Isoparametric interface element

The fracture simulation by the discrete approach requires the addition of interface elements (cohesive elements) to a conventional finite element mesh in order to provide the domain with potential crack paths. In this regard, this section begins by presenting the formulation of the isoparametric interface element used in this work. Similar formulations are given in Beer (1985); Plesha et al. (1989); Hird and Kwok (1989); Roy and Dodds (2001); Jin et al. (2003). Later, this section shows the algorithm used to add interface elements between all bulk elements in a conventional mesh.

### 2.1 Formulation of the isoparametric interface element

Figure 2 shows a four-node interface element with length $L$ and zero thickness. The figure at left shows the element located at the global coordinate system $xy$ while the figure at right shows the element in the parametric space $\xi$. Although the figure shows an element with linear interpolation, more nodes can be considered to get, for example, quadratic interpolation, which allows the simulation of curved contacts.

The displacements vector of a four-node interface element is given by:

$$\mathbf{U} = \langle u_{x1},\ u_{y1},\ u_{x2},\ u_{y2},\ u_{x3},\ u_{y3},\ u_{x4},\ u_{y4} \rangle . \qquad (1)$$

Since the faces of an interface element may have different displacement fields, **U** can be used to calculate a vector with relative displacements $\mathbf{w} = \langle w,\ v \rangle$ in the local reference (normal and tangent displacements) at a

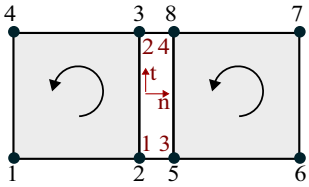



**Fig. 3** Global (black) and local (red) numbering for the interface element nodes

point $\xi$ (in parametric coordinates) along the interface element as:

$$\mathbf{w}(\xi) = \mathbf{T}(\xi)\mathbf{N}(\xi)\mathbf{U}, \tag{2}$$

where $\mathbf{N}(\xi)$ is an interpolation matrix based on shape functions and $\mathbf{T}(\xi)$ is a rotation matrix used to express global displacements in the local reference. The following equation shows matrix $\mathbf{T}$, where $n_i$ and $t_i$ are the components of the normal and tangential unitary vectors ($\hat{\mathbf{n}}$ and $\hat{\mathbf{t}}$), respectively:

$$\mathbf{T} = \begin{bmatrix} n_1 & n_2 \\ t_1 & t_2 \end{bmatrix}. \tag{3}$$

The tangential unitary vector is easily calculated as $\hat{\mathbf{t}} = \frac{\mathbf{J}^T}{||\mathbf{J}^T||}$, where $\mathbf{J}$ is the Jacobian matrix, i.e. $\left[\frac{\partial x}{\partial \xi} \;\; \frac{\partial y}{\partial \xi}\right]$, of a line element considering the nodes from one face of the interface. The normal vector, in turn, is given by $\mathbf{n} = \langle J_2, -J_1 \rangle$, where $J_1$ and $J_2$ are the components of $\mathbf{J}$. Figure 3 shows an example of two bulk elements linked by an interface element. In this case, the interface connectivity is given by the nodes 2, 3, 5, and 8. Note that the first nodes (2 and 3) are numbered following the same sequence as the first neighbor element (at left). Regarding the interpolation matrix $\mathbf{N}$, it is given by:

$$\mathbf{N} = \begin{bmatrix} -N_1 & 0 & -N_2 & 0 & N_1 & 0 & N_2 & 0 \\ 0 & -N_1 & 0 & -N_2 & 0 & N_1 & 0 & N_2 \end{bmatrix}, \tag{4}$$

where $N_1$ and $N_2$ are the same shape functions as those from a two-node line element considering the interface first two nodes.

Equation 2 can be rewritten incrementally as below where the symbol $\xi$ was dropped for clarity:

$$\Delta\mathbf{w} = \mathbf{T}\,\mathbf{N}\,\Delta\mathbf{U}. \tag{5}$$

The stress state at each point along the interface element is represented by the vector $\boldsymbol{\sigma} = \langle \sigma_n, \tau \rangle$, where $\sigma_n$ and $\tau$ represent the normal and shear stresses, respectively: the first one in the $\hat{\mathbf{n}}$ direction and the second one in the $\hat{\mathbf{t}}$ direction. The increments in stress components $\Delta\boldsymbol{\sigma}$ are related to the increment of relative displacements $\Delta\mathbf{w}$ by means of a constitutive matrix $\mathbf{D}$:

$$\Delta\boldsymbol{\sigma} = \mathbf{D}\,\Delta\mathbf{w}. \tag{6}$$

Using a conventional finite element procedure, as the virtual work method, we can quickly arrive at the following equation for the stiffness matrix of an interface element in two dimensions:

$$\mathbf{K} = \int_L \mathbf{B}^T\mathbf{D}\mathbf{B}\,dL, \tag{7}$$

where $\mathbf{B} = \mathbf{T}\mathbf{N}$. Matrix $\mathbf{K}$ can be calculated using numerical integration; for this purpose, a number of integration points must be defined according to the degree of the integrand. Equation 7 can be easily expanded to three-dimensional analyses by conveniently modifying matrices $\mathbf{N}$, $\mathbf{T}$ and $\mathbf{D}$.

### 2.2 Computation of a representative length of bulk elements

The simple addition of cohesive elements to a conventional mesh would make the domain more flexible due to additional degrees of freedom and interface elements deformability. To reduce mesh dependency and to obtain convergent results in h-refinement analyses, the stiffness from interface elements needs to be related to the size of neighbor bulk elements, e.g. a representative length. We define this representative length $h$ as the average bulk element length in the perpendicular direction to the cohesive element. Consider, for example, the meshes in Fig. 4 and assume that all bulk elements are linked by cohesive elements. In the first case, the representative length is given by the elements' width, since all bulk elements are equal. In the second case, the representative length for the cohesive element between the two shaded elements will be the average between the widths $h_1$ and $h_2$. In the last case, the value of $h$ for the shaded elements is computed considering two rectangular elements with equivalent areas, $A_1$ and $A_2$. This allows to calculate two equivalent widths: $h_1 = A_1/L$ and $h_2 = A_2/L$. Finally, the representative length is given by the average value of $h_1$ and $h_2$:

$$h = \frac{A_1 + A_2}{2L}, \tag{8}$$

Since this formula satisfies previous cases, it can be used for any pair of bulk elements.

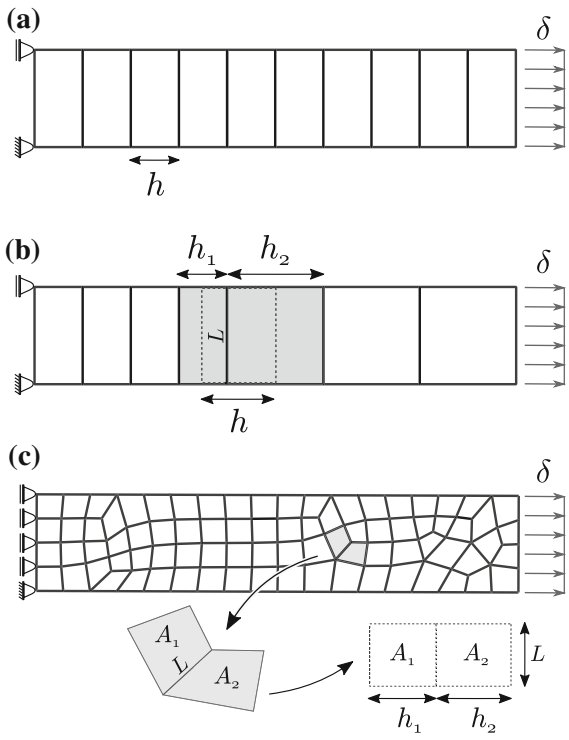



**Fig. 4** Computation of the representative length of bulk elements: **a** in a mesh with equal rectangular elements; **b** in a mesh with rectangular elements with different sizes; **c** in a unstructured mesh

### 2.3 Addition of cohesive elements to a finite element mesh

This procedure deals with the insertion of cohesive elements between bulk ones in a finite element mesh as is shown in Fig. 5. This is not a simple task since the faces of an interface element have the same set of coordinates, which makes them difficult to distinguish. Existing cohesive element formulations use algorithms of this kind to insert interface elements among all element boundaries (see e.g. Paulino et al. 2008; Nguyen 2014; Jiang and Meng 2018; Tabiei and Zhang 2017). In this section, we aim to provide a detailed explanation of the procedure used in this work. This procedure is presented in Algorithm 1 and can be explained in three main steps. These steps are described in the context of object oriented programming, thus we assume that a mesh is described in terms of node and element objects. We also assume that each element object contains a list of pointers to its corresponding nodes. The referred steps are:

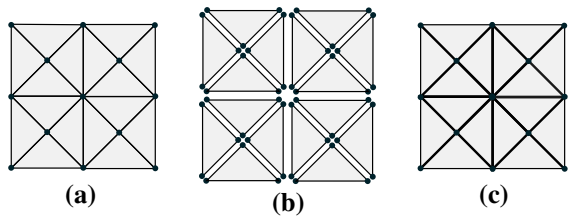



**Fig. 5** Generation of mesh with interface elements: **a** initial mesh; **b** elements detachment (new nodes are generated and elements are detached); **c** interface elements are generated by connecting matching edges

1. Elements detachment: In this step all bulk elements in the mesh are detached from its neighbors to avoid the sharing of nodes (Fig. 5b). For this purpose, at each element, the nodal pointers are replaced with pointers to copies of the original nodes. In other words, for each element, new nodes are created and linked in substitution to the original ones. This procedure provides an entirely new set of nodes for the mesh thus the original nodes can be discarded.
2. Pairing of edges: For each element, a list of boundary edges is generated. Each edge contains a list of corresponding nodal pointers. Later, a hash value is calculated for each edge on all elements. The purpose of using a hash value is to easy checking whether two edges have the same location, which is the case at the common side of two neighbor elements. Thus, a hash function must be defined in a way that returns the same value for two edges placed at the same location. When two edges with the same hash value are detected, an edge pair object is generated and stored into a list.
3. Generation of interface elements: For each edge pair found in the previous step, a new interface element is generated. The connectivity of an interface element is given by a list with nodal pointers taken from the corresponding edges. Finally, the mesh is updated to include the generated interface elements.

## 3 Constitutive modeling

In this work, we propose a plasticity model to simulate the mechanical behavior of cohesive elements. The model is defined by the Coulomb law with adherence and, according to Lens et al. (2009), we define the yield surface as:

**Algorithm 1:** Interface elements addition algorithm.

**Input**: A conventional mesh called *Mesh*
**Output**: The updated version of *Mesh* including interface elements located between 2D elements

*! Step 1: detachment of elements*
get a list *Elements* with all elements from *Mesh*

**foreach** *elem* in *Elements* **do**
  copy the data from the nodes in *elem* to new node objects
  replace nodal pointers in *elem* with pointers to corresponding copies
  discard original node objects

*! Step 2: pairing of edges with same hash value*
*EdgePairs* = [ ] *! A list for pairs of matching edges*
*Edges* = [ ] *! A list for all edges*
**foreach** *elem* in *Elements* **do**
  create a list *ElemEdges* with all edges of *elem*
  **forall the** edge *e* in *ElemEdges* **do**
    calculate a hash value for *e*
    **if** an edge $\bar{e}$ from *Edges* has the same hash value as edge *e* **then**
      add the pair $(e, \bar{e})$ to *EdgePairs*
      remove *e* from *Edges*
    **else**
      add *e* to *Edges*

*! Step 3: generation of interface elements*
*Interfaces* = [ ] *! List for newly generated interface elements*
**foreach** *pair* in *EdgePairs* **do**
  *INodes* = [ ] *! A list with pointers to nodes for an interface element*
  **foreach** edge *e* in *pair* **do**
    get all nodes from *e* and add its pointers to *INodes*
  generate an interface element *I* with connectivity based on *INodes*
  add *I* to *Interfaces*

update the list of nodes in *Mesh* with all nodes found in *Interfaces*
update the list of elements in *Mesh* with the elements in *Interfaces*
renumerate all nodes and elements in *Mesh*

$$f(\boldsymbol{\sigma}, \sigma_{max}) = |\tau| + (\sigma_n - \sigma_{max}) \tan\phi = 0, \tag{9}$$

where $\sigma_n$ and $\tau$ are, respectively, the normal and shear stresses in the crack plane while $\sigma_{max}$ is the internal softening variable, i.e. the material tensile strength, and $\phi$ is the material friction angle. Figure 6 shows the yield surface (solid line) in the $\sigma_n - \tau$ space. In the figure $\sigma^0_{max}$ and $\tau^0_{max}$ represent the initial tensile and shear strength of concrete, respectively. The proposed model

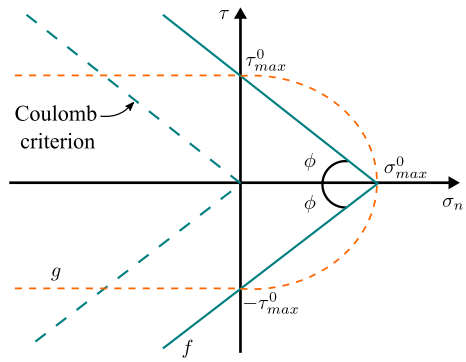



**Fig. 6** Yield function (solid line) and plastic potential function (dashed line) after Lens et al. (2009)

uses a non associated flow rule. The plastic potential function is shown in Fig. 6 as an orange dashed line and its equation is given by:

$$g(\boldsymbol{\sigma}, \sigma_{max}) = \begin{cases} \tau^2 - \tau^2_{max} = 0 \quad \text{for} \quad \sigma_n \leq 0 \\ \sigma_n^2 \tan^2\phi + \tau^2 - \tau^2_{max} = 0 \quad \text{for} \quad \sigma_n > 0 \end{cases} \tag{10}$$

The internal variable $\sigma_{max}$ defines the location of the yield surface aiming to simulate the loss of strength of concrete due to cracking. This variable is a function of the effective plastic relative displacement defined as:

$$u_p = \int_0^t ||\dot{\mathbf{w}}_p||\, dt = \int_0^t \sqrt{\dot{w}_p^2 + \dot{v}_p^2}\, dt, \tag{11}$$

where $\dot{w}_p$ and $\dot{v}_p$ are the rate of normal and shear plastic relative displacements, respectively, and $\dot{\mathbf{w}}_p = \langle \dot{w}_p, \dot{v}_p \rangle$. There are several softening curves available in the literature to relate $\sigma_{max}$ and $u_p$. In this work, two softening curves are investigated: a bilinear and an exponential function, the later proposed by Hordijk (1992). Both curves are presented in Fig. 7 where $u_c$ represents the critical crack opening. As seen in these curves, $\sigma_{max}$ reduces when $u_p$ increases; this causes the displacement of the yield surface as shown in Fig. 8.

### 3.1 Bilinear softening curve

This curve presents two linear segments as shown in Fig. 7a where $(u_s, \sigma_s)$ represent the point where the curve slope changes. Furthermore, $u_c$ represents the

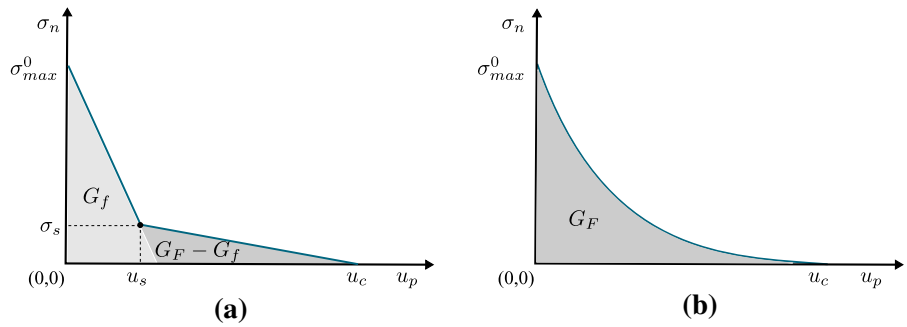



**Fig. 7** Softening curve of concrete: **a** bilinear softening curve; **b** Hordijk softening curve

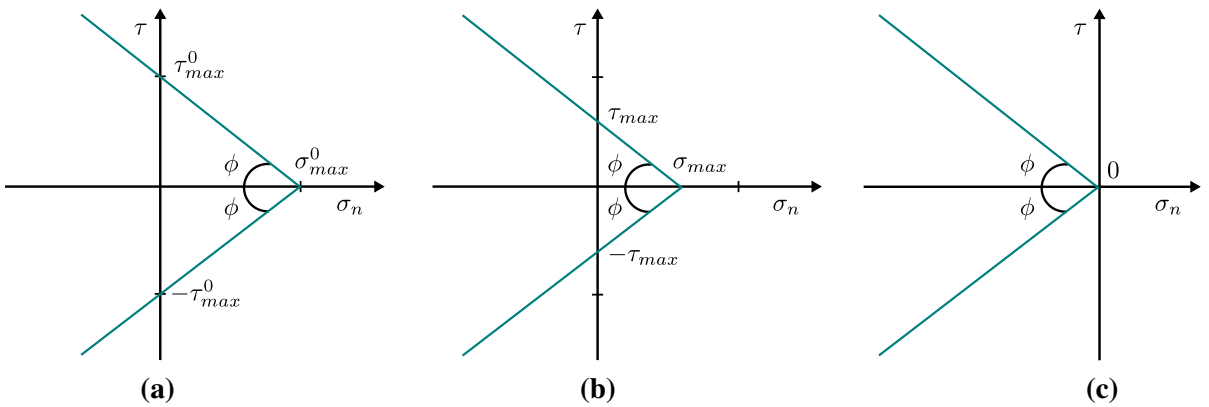



**Fig. 8** Yield function evolution: **a** initial location ($u_p = 0$); **b** intermediary location ($0 < u_p < u_c$); c) final location ($u_p \geq u_c$)

critical crack opening when $\sigma_{max}$ is equal to zero. Bilinear curves are considered as reasonable approximations; se e.g. (Wittmann et al. 1988; Yang and Chen 2005; Roesler et al. 2007; Park et al. 2008; Park and Paulino 2013). Although there is no agreement about the location of the point $(u_s, \sigma_s)$, in this work $\sigma_s$ is assumed as $0.25\,\sigma^0_{max}$ as proposed by Wittmann et al. (1988). The terms $u_s$ and $u_c$ are calculated in such a way that the area below the curve is equal to the fracture energy $G_F$:

$$u_c = \frac{5\,G_F}{\sigma^0_{max}} \quad \text{and} \tag{12}$$

$$u_s = 0.15 u_c. \tag{13}$$

When the initial fracture energy $G_f$ is also available, $u_c$ and $u_s$ are calculated as:

$$u_c = \frac{8\,G_F - 6\,G_f}{\sigma^0_{max}} \quad \text{and} \tag{14}$$

$$u_s = \frac{1.5\,G_f}{\sigma^0_{max}}. \tag{15}$$

Later, $\sigma_{max}$ as a function of $u_p$ varies according to:

$$\sigma_{max}(u_p) = \begin{cases} \sigma^0_{max} - u_p \left(\dfrac{\sigma^0_{max} - \sigma_s}{u_s}\right) & \text{for} \quad 0 < u_p \leq u_s \\ \dfrac{u_c\,\sigma_s}{u_c - u_s} - u_p \left(\dfrac{\sigma_s}{u_c - u_s}\right) & \text{for} \quad u_s < u_p \leq u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases} \tag{16}$$

The rate of change of $\sigma_{max}$ with respect to $u_p$ is given by:

$$\frac{\partial \sigma_{max}}{\partial u_p} = \begin{cases} -\left(\dfrac{\sigma^0_{max} - \sigma_s}{u_s}\right) & \text{for} \quad 0 < u_p \leq u_s \\ -\left(\dfrac{\sigma_s}{u_c - u_s}\right) & \text{for} \quad u_s < u_p \leq u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases} \tag{17}$$

### 3.2 Hordijk's softening curve

This softening curve is given by an exponential function Hordijk (1992) and depends only on two parameters,

$\sigma^0_{max}$ and $u_c$ . As in the bilinear curve, $u_c$ is calculated in order to provide an area under the curve equal to $G_F$. The Hordijk's softening curve, using the coefficients proposed by Zivaljic et al. (2014), is written as:

$$\sigma_{max}(u_p) = \begin{cases} \left[\left(1+27\left(\frac{u_p}{u_c}\right)^3\right) e^{-6.93\frac{u_p}{u_c}} - 28\frac{u_p}{u_c}\, e^{-6.93}\right]\sigma^0_{max} & \text{for} \quad u_p \le u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases} \tag{18}$$

After integration of Eq. 18 from 0 to $u_c$ and equating to $G_F$ we get:

$$u_c = \frac{G_F}{0.1947\,\sigma^0_{max}}. \tag{19}$$

The rate of change $\frac{\partial \sigma_{max}}{\partial u_p}$, required in the constitutive model formulation, is given by:

$$\frac{\partial \sigma_{max}}{\partial u_p} = \begin{cases} \left(\dfrac{81\,u_p^2\, e^{\frac{-6.93\,u_p}{u_c}}}{u_c^3} - \dfrac{6.93\left(1+\frac{27\,u_p^3}{u_c^3}\right) e^{\frac{-6.93\,u_p}{u_c}}}{u_c} - \dfrac{0.02738402432}{u_c}\right)\sigma^0_{max} & \text{for} \quad u_p \le u_c \\ 0 & \text{for} \quad u_p > u_c \end{cases} \tag{20}$$

The value of $\alpha$ should be appropriately chosen in order to provide negligible displacement values for $w_{e(max)}$ and $v_{e(max)}$ at the elastic regime where no crack openings are expected. Too high values for $\alpha$ may provide high stiffness coefficients and consequently an ill-conditioned system of equations. In this work $\alpha = 5$ provided satisfactory results for all examples. Bigger values for $\alpha$ may provide a singular system of equations in some cases.

For the elastoplastic regime, the constitutive matrix $\mathbf{D}_{ep}$ is formulated based on the conventional plasticity theory. According to the additive decomposition, the increment of relative displacements at a point along the cohesive element is expressed as the sum of the elastic and plastic components:

### 3.3 Constitutive matrix

This section presents the elastic and elastoplastic constitutive matrix $\mathbf{D}$ that relates the relative displacement increment vector $\dot{\mathbf{w}}$ with the stress increment vector $\dot{\boldsymbol{\sigma}}$ at an integration point of a cohesive element. At the elastic regime, the constitutive matrix is given by:

$$\mathbf{D}_e = \begin{bmatrix} k_n & 0 \\ 0 & k_t \end{bmatrix}, \tag{21}$$

where $k_n$ and $k_t$ are, respectively, related to the normal and tangential stiffness in the crack plane. They are defined as:

$$k_n = E\frac{\alpha}{h} \quad \text{and} \quad k_t = G\frac{\alpha}{h}, \tag{22}$$

where $h$ is the representative length from bulk elements (described in item 2.2), $E$ is the Young's modulus, $G$ is the shear modulus and $\alpha$ is a factor that controls the elastic relative displacements. The $\alpha$ factor can be used to define the maximum elastic relative displacements allowed in the normal $w_{e(max)}$ and tangential $v_{e(max)}$ directions according to:

$$w_{e(max)} = \frac{\sigma^0_{max}}{\alpha E}h \quad \text{and} \quad v_{e(max)} = \frac{\tau^0_{max}}{\alpha G}h. \tag{23}$$

$$\dot{\mathbf{w}} = \dot{\mathbf{w}}_e + \dot{\mathbf{w}}_p. \tag{24}$$

Notice that plastic displacements are not reversible. Thus in an unloading process, only part of the displacements are reversible. Consequently, once plastic displacements occur, total displacements are no longer proportional to stresses.

The increment of plastic relative displacements is calculated by the flow rule according to:

$$\dot{\mathbf{w}}_p = \dot{\lambda}\,\mathbf{r}, \tag{25}$$

where $\mathbf{r}$ is a vector normal to the potential surface $g$ and $\dot{\lambda}$ is the plastic multiplier. Thus the increment of the effective plastic relative displacement is calculated as:

$$\dot{u}_p = ||\dot{\mathbf{w}}_p|| = \dot{\lambda}\,||\mathbf{r}||. \tag{26}$$

After multiplying Eq. 24 by $\mathbf{D}_e$ and considering $\dot{\boldsymbol{\sigma}} = \mathbf{D}_e\dot{\mathbf{w}}_e$ and Eq. 25 we get the stress increment vector as:

$$\dot{\boldsymbol{\sigma}} = \mathbf{D}_e\,\dot{\mathbf{w}} - \dot{\lambda}\,\mathbf{D}_e\,\mathbf{r}. \tag{27}$$

On the other hand, according to Eq. 9, the consistency condition is given by:

$$\dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \sigma_{max}}\dot{\sigma}_{max} = 0, \tag{28}$$

which can be expanded as:

$$\dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \left(\mathbf{D}_e \, \dot{\mathbf{w}} - \dot{\lambda} \, \mathbf{D}_e \, \mathbf{r}\right) + \frac{\partial f}{\partial \sigma_{max}} \frac{\partial \sigma_{max}}{\partial u_p} \dot{u}_p = 0. \tag{29}$$

Using Eq. 26 and isolating the increment of the plastic multiplier, we get:

$$\dot{\lambda} = \frac{\dfrac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \mathbf{D}_e \, \dot{\mathbf{w}}}{\dfrac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \mathbf{D}_e \, \mathbf{r} - \dfrac{\partial f}{\partial \sigma_{max}} \dfrac{\partial \sigma_{max}}{\partial u_p} ||\mathbf{r}||} = \frac{\mathbf{v} \cdot \mathbf{D}_e \, \dot{\mathbf{w}}}{\mathbf{v} \cdot \mathbf{D}_e \, \mathbf{r} - y \, m ||\mathbf{r}||}, \tag{30}$$

where $\mathbf{v} = \frac{\partial f}{\partial \boldsymbol{\sigma}}$, $y = \frac{\partial f}{\partial \sigma_{max}}$ and $m = \frac{\partial \sigma_{max}}{\partial u_p}$. Later, by substituting Eq. 30 into Eq. 27, it is possible to find the elastoplastic constitutive matrix $\mathbf{D}_{ep}$ as:

$$\mathbf{D}_{ep} = \mathbf{D}_e - \frac{\mathbf{D}_e \, \mathbf{r} \, \mathbf{v}^T \, \mathbf{D}_e}{\mathbf{v}^T \, \mathbf{D}_e \, \mathbf{r} - y \, m ||\mathbf{r}||}. \tag{31}$$

The required derivatives for the calculation of the elastoplastic constitutive matrix are given by:

$$y = -\tan \Phi, \tag{32}$$

$$\mathbf{v} = \left\langle \frac{\partial f}{\partial \sigma_n}, \; \frac{\partial f}{\partial \tau} \right\rangle = \langle \tan \Phi, \; \text{sgn}\,(\tau) \rangle \,, \tag{33}$$

$$\mathbf{r} = \left\langle \frac{\partial g}{\partial \sigma_n}, \; \frac{\partial g}{\partial \tau} \right\rangle = \left\langle 2\sigma_n \tan^2 \phi, \; 2\tau \right\rangle \quad \text{for} \quad \sigma_n > 0, \tag{34}$$

$$\mathbf{r} = \left\langle \frac{\partial g}{\partial \sigma_n}, \; \frac{\partial g}{\partial \tau} \right\rangle = \langle 0, \; 2\tau \rangle \quad \text{for} \quad \sigma_n < 0. \tag{35}$$

### 3.4 Stress update

The purpose of the stress update process is to find the stress vector increment $\Delta\boldsymbol{\sigma}$ for a given increment of relative displacements $\Delta\mathbf{w}$. Initially, a trial stress vector is found by using the elastic constitutive matrix as:

$$\boldsymbol{\sigma}^{tr} = \boldsymbol{\sigma}^n + \mathbf{D}_e \, \Delta\mathbf{w}, \tag{36}$$

where the superscript $n$ represents the current step.

Aiming to find if the increment of relative displacements leads the stress state to the elastic or elastoplastic regime, the signal of the yield function is evaluated at $\boldsymbol{\sigma}^{tr}$. Thus, if $f(\boldsymbol{\sigma}^{tr}, \sigma^n_{max}) \leq 0$ the increment is elastic and the new stress state is $\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{tr}$. This occurs during initial loadings and unloading/reloading processes. In the case of $f(\boldsymbol{\sigma}^{tr}, \sigma^n_{max}) > 0$, the increment is elastoplastic and consequently the internal variable $\sigma_{max}$ will decrease. This implies that the yield surface will displace to the left at step $n+1$. Also, the stress state given by $\boldsymbol{\sigma}^{tr}$ should return to the yield surface following a direction perpendicular to the potential surface $g$ as shown in Fig. 9. The required equations to update $u_p$, $\sigma_{max}$ and $\boldsymbol{\sigma}$ for a given increment of relative displacements $\Delta\mathbf{w}$ in the elastoplastic regime are:

$$u_p^{n+1} = u_p^n + \int_{t_n}^{t_{n+1}} \dot{u}_p \, dt = u_p^n + \Delta\lambda \, ||\mathbf{r}^{n+1}||, \tag{37}$$

$$\sigma_{max}^{n+1} = \sigma_{max}^n + \frac{\partial \sigma_{max}}{\partial u_p} \Delta\lambda \, ||\mathbf{r}^{n+1}||, \tag{38}$$

$$\boldsymbol{\sigma}^{n+1} = \boldsymbol{\sigma}^{tr} - \Delta\lambda \, \mathbf{D}_e \, \mathbf{r}^{n+1}. \tag{39}$$

Equation 39 was obtained by substituting $\Delta\mathbf{w}^{n+1} = \Delta\mathbf{w}_e^{n+1} + \Delta\mathbf{w}_p^{n+1}$ and $\Delta\mathbf{w}_p^{n+1} = \Delta\lambda \, \mathbf{r}^{n+1}$ into Eq. 36. Note that, according to Eq. 38 and the fact that $\frac{\partial \sigma_{max}}{\partial u_p} < 0$, $\Delta\lambda$ must be a positive number in order to be consistent with the loss of strength due to cracking.

From the plastic potential function $g$ (Fig. 6) we verify that $\text{sgn}\,(\sigma_n^{n+1}) = \text{sgn}\,(\sigma_n^{tr})$. This implies that the signal of $\sigma_n^{tr}$ can be used to choose the right expression for $\mathbf{r}$ to be employed to find the stresses at step $n+1$ (see Eqs. 34 and 35 ), thus we have:

$$\langle \sigma_n^{n+1}, \; \tau^{n+1} \rangle = \left\langle \frac{\sigma_n^{tr}}{1 + 2\Delta\lambda \, k_n \tan^2 \phi}, \; \frac{\tau^{tr}}{1 + 2\Delta\lambda \, k_s} \right\rangle \qquad \text{for} \quad \sigma_n^{tr} > 0, \tag{40}$$

$$\langle \sigma_n^{n+1}, \; \tau^{n+1} \rangle = \left\langle \sigma_n^{tr}, \; \frac{\tau^{tr}}{1 + 2\Delta\lambda \, k_s} \right\rangle \quad \text{for} \quad \sigma_n^{tr} \leq 0. \tag{41}$$

To find $\Delta\lambda$, needed to update $u_p$, $\sigma_{max}$ and $\boldsymbol{\sigma}$, let's consider the consistency condition at the step $n+1$:

$$f(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1}) = |\tau^{n+1}| + (\sigma_n^{n+1} - \sigma_{max}^{n+1}) \tan \phi = 0. \tag{42}$$

After substituting Eqs. 38, 40 and 41 in the equation above we get a nonlinear equation in terms of $\Delta\lambda$ and can be solved by applying a root-finding numerical method. In this case we use the Newton method, thus:

$$\Delta\lambda_{i+1} = \Delta\lambda_i - \frac{f(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1})_i}{f'(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1})_i}, \tag{43}$$

where $i$ indicates the $i$-th iteration, $\Delta\lambda_0$ is an initial guess (e.g. $\Delta\lambda_0 = 0$) and $f'(\boldsymbol{\sigma}^{n+1}, \sigma_{max}^{n+1})$ can be expressed as:

$$\frac{\partial f^{n+1}}{\partial \Delta\lambda} = \frac{\partial f^{n+1}}{\partial \boldsymbol{\sigma}^{n+1}} \cdot \frac{\partial \boldsymbol{\sigma}^{n+1}}{\partial \Delta\lambda} + \frac{\partial f^{n+1}}{\partial \sigma_{max}^{n+1}} \frac{\partial \sigma_{max}^{n+1}}{\partial \Delta\lambda}. \tag{44}$$

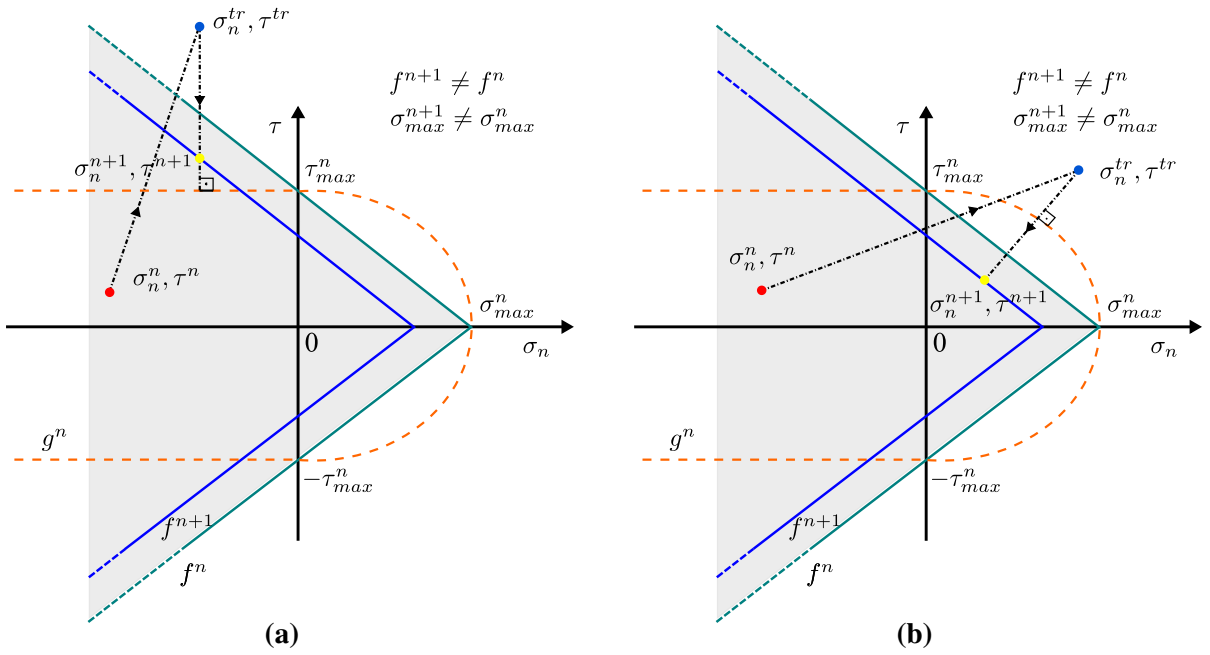



**Fig. 9** Stress update in the elastoplastic regime: **a** stress return to the yield surface when $\sigma_n^{tr} < 0$; **b** stress return to the yield surface when $\sigma_n^{tr} > 0$

To compute the yield function derivative, the following derivatives are required, where the superscript $n+1$ was dropped for clarity:

$$\frac{\partial \boldsymbol{\sigma}}{\partial \Delta\lambda} = \left\langle \frac{-2k_n \tan^2 \Phi\, \sigma_n^{tr}}{(1+2\Delta\lambda k_n \tan^2 \Phi)^2}, \frac{-2k_s \tau^{tr}}{(1+2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} > 0, \tag{45}$$

$$\frac{\partial \boldsymbol{\sigma}}{\partial \Delta\lambda} = \left\langle 0, \frac{-2k_s \tau^{tr}}{(1+2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} \leq 0, \tag{46}$$

$$\frac{\partial \mathbf{r}}{\partial \Delta\lambda} = \left\langle \frac{-4k_n \tan^4 \Phi\, \sigma_n^{tr}}{(1+2\Delta\lambda k_n \tan^2 \Phi)^2}, \frac{-4k_s \tau^{tr}}{(1+2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} > 0, \tag{47}$$

$$\frac{\partial \mathbf{r}}{\partial \Delta\lambda} = \left\langle 0, \frac{-4k_s \tau^{tr}}{(1+2\Delta\lambda k_s)^2} \right\rangle \quad \text{for } \sigma_n^{tr} \leq 0, \tag{48}$$

$$\frac{\partial \boldsymbol{\sigma}_{max}}{\partial \Delta\lambda} = \frac{\sigma_{max}}{u_p} \left( ||\mathbf{r}|| + \Delta\lambda \hat{\mathbf{r}} \cdot \frac{\partial \mathbf{r}}{\partial \Delta\lambda} \right) \tag{49}$$

### 3.5 Plastic update when $u_p \geq u_c$

This procedure is one key aspect of the proposed constitutive model. When a crack achieves the critical opening ($u_p \geq u_c$) there is no more tensile strength, thus $\sigma_{max} = 0$. This causes the potential surface to collapse into a line and its derivatives become undetermined at the origin. To deal with this situation, we start by checking whether, at step $n+1$, the crack will be open enough to be unable to transfer stresses, that is $w^{n+1} \geq u_c$. We use $u_c$ and not zero in the condition since we assume that, due to surface irregularities, the crack is able to transfer compressive and shear stresses if their faces are not completely separated. Thus, if $w^{n+1} \geq u_c$ then the normal and shear stresses at step $n+1$ should be equal to zero. Figure 10 shows the stress update process for two possible states at step $n$: when the crack faces are separated ($w^n \geq u_c$) and when the crack faces are in contact ($w^n < u_c$, e.g. by previous unloading). In both cases, stresses at step $n+1$ are set to zero and Eq. 39 is rewritten as:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \sigma_n^{tr} \\ \tau^{tr} \end{pmatrix} - \Delta\lambda \begin{bmatrix} k_n & 0 \\ 0 & k_t \end{bmatrix} \begin{pmatrix} r_1^{n+1} \\ r_2^{n+1} \end{pmatrix}. \tag{50}$$

Later, vector $\mathbf{r}^{n+1}$ can be expressed as:

$$\mathbf{r}^{n+1} = \frac{1}{\Delta\lambda} \begin{pmatrix} \frac{\sigma_n^{tr}}{k_n} \\ \frac{\tau^{tr}}{k_t} \end{pmatrix}. \tag{51}$$

By assuming $\mathbf{r}^{n+1}$ as an unitary vector, $\Delta\lambda$ is calculated as the norm of vector $\langle \frac{\sigma_n^{tr}}{k_n}, \frac{\tau^{tr}}{k_t} \rangle$ and the stress update

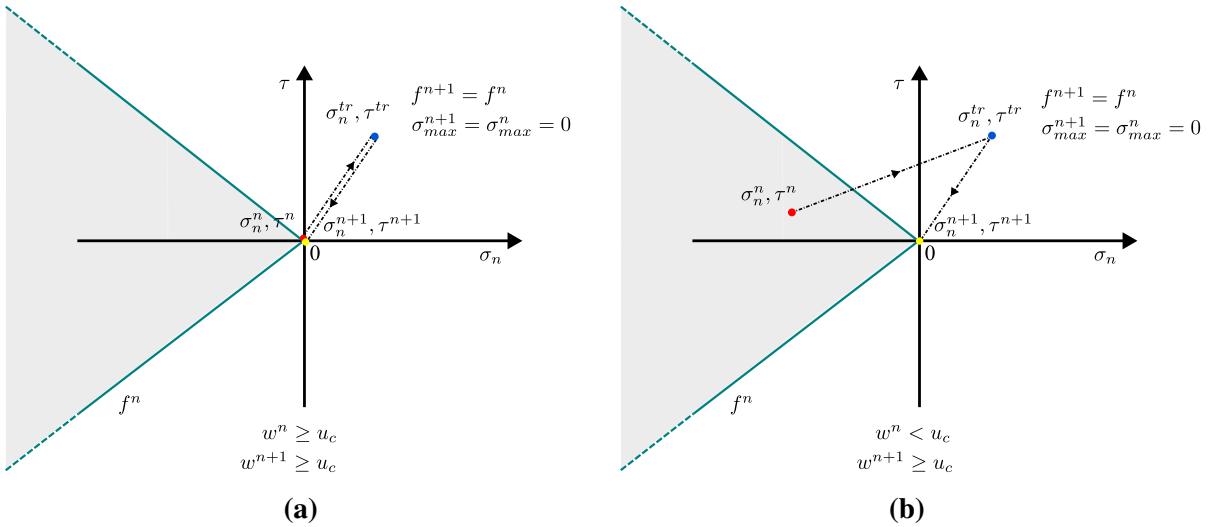



**Fig. 10** Stress update for the condition $u_p^{n+1} \geq u_c$ : **a** the crack faces are separated at step $n$; **b** the crack faces are in contact at step $n$

process can be performed using Eq. 39. Since the crack is open at step $n+1$, it should be considered $\mathbf{D}_{ep} = \mathbf{0}$ for the next load step while computing the element stiffness matrix.

### 3.6 Crack opening during unloading and reloading

During the elastoplastic loading, the crack opening increment is given by the sum of its elastic and plastic components, $\Delta w = \Delta w_e + \Delta w_p$, where, according to the flow rule, $\Delta w_p = \Delta\lambda\, r_1$. Thus, the total plastic opening at step $n+1$ can be stated as $w_p^{n+1} = w_p^n + \Delta\lambda\, r_1$. In the case of subsequent unloading and reloading events, $w_p$ remains unaltered in the same way as $u_p$.

Figure 11 shows an example of a normal stress versus crack opening path, which is based on the application of controlled displacements and includes unloading and reloading branches. The elastic and plastic components of the crack opening at point $A$ are specified by $w_e^A$ and $w_p^A$, respectively. The unloading event from point $A$ follows a linear elastic path to point $B$, where $\sigma_n$ is zero. Subsequent closing keeps $\sigma_n$ unchanged from point $B$ to $C$. Similarly, during reloading, the normal stress is kept constant, from point $C$ to $D$, until the opening $w$ reaches the value of $w_p^A$, then the path continues in a linear elastic fashion. Once the stress state reaches the yield function, which is compatible with point $E$, the elastoplastic behavior is recovered.

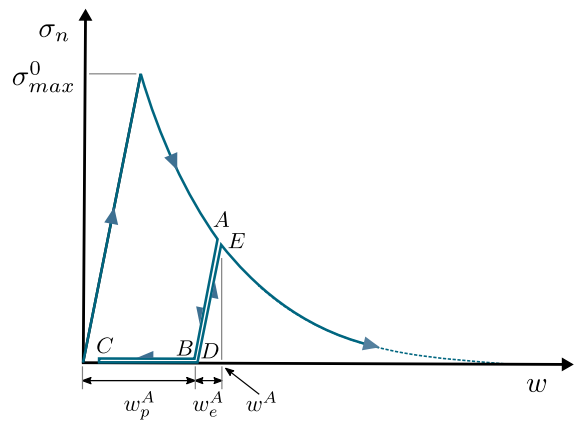



**Fig. 11** Normal stress versus crack opening in a typical unloading/reloading process

## 4 Numerical simulations

This section presents two validation tests and five application examples of the proposed model. The validation tests aim to verify the cohesive constitutive model behavior in extension and shear simulations. Also, a mesh objectivity test is presented to verify that the application of the proposed formulation minimizes mesh dependency. On the sequence, five application

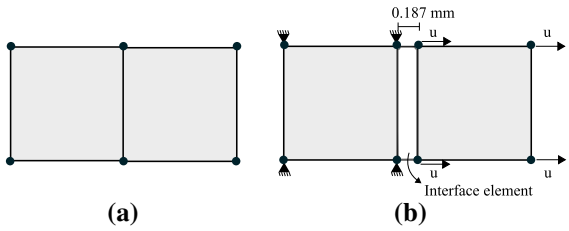
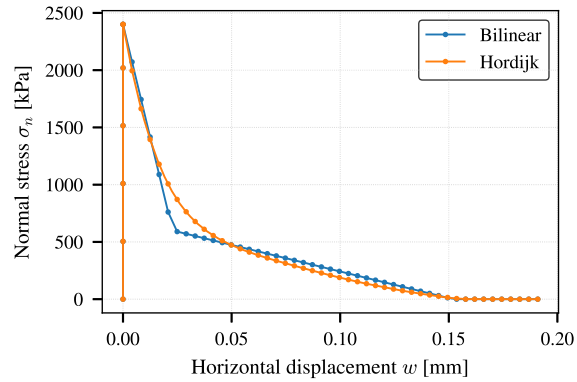
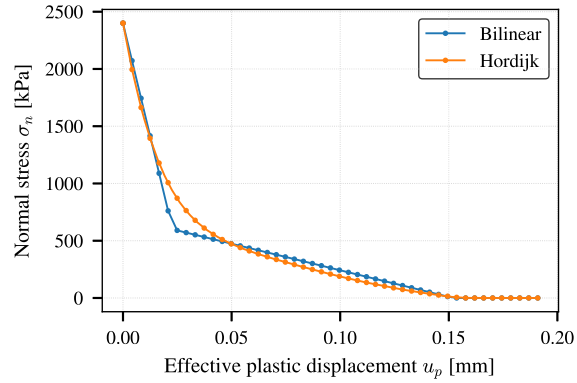



**Fig. 12** Extension test; **a** initial configuration; **b** deformed configuration

examples deal with fracture simulation of experimental concrete specimens. The examples study the effect of softening curves (bilinear vs Hordijk's), type of elements (triangular vs quadrilateral), element interpolation (linear or bilinear vs quadratic) and mesh refinement. For this purpose, a total of 28 analyses were carried out. In all cases, planes stress conditions are considered since specimens are fairly thin and no tractions are applied perpendicular to the $xy$ plane. Also, linear-elastic properties are considered for all bulk elements. Although it is possible to place cohesive elements only in specific regions, we decided to place them among all elements boundaries as a means to verify that cracks only happen where expected. The numerical simulations were performed using a finite element solver with automatic load/displacement stepping.




**(a)** Horizontal displacement vs normal stress




**(b)** Effective plastic displacement vs normal stress

**Fig. 13** Extension test results

## 4.1 Validation tests

In this topic, we present two validations tests for the proposed constitutive model in order to verify the results when a cohesive element is submitted to extension and shear. Also, a simple mesh objectivity test is presented.

### *4.1.1 Extension and shear tests*

For the extension test, a mesh with two four-node quadrilateral elements and one four-node cohesive element was used. The mesh and boundary conditions are presented in Fig. 12 where the sides of the quadrilateral elements are equal to 10 mm. As seen in Fig. 12b, the bulk element at left was constrained in all directions while the bulk element at right was subjected to rigid body movement; thus, all deformations are constrained to the cohesive element. The bilinear and Hordijk's curves were used to model softening. The required parameters for these models are presented in Table 1 where $\nu$ is the Poisson ratio and $\sigma^0_{max}$ is the material tensile strength. Parameters $u_s$ and $u_c$ were obtained considering a concrete fracture energy $G_F = 73$ N/m.

In this test, horizontal displacements equal to $u = 0.187$ mm were applied in 50 increments. During the analysis, stresses and relative displacements were

**Table 1** Material parameters for the cohesive model used in the validation tests

| Softening curve | $E$ (GPa) | $\nu$ | $\tan(\phi)$ | $\alpha$ | $\sigma^0_{max}$ (MPa) | $u_s$ (mm) | $u_c$ (mm) |
|---|---|---|---|---|---|---|---|
| Bilinear | 27.0 | 0.20 | 1.4 | 5 | 2.40 | 0.0228 | 0.152 |
| Hordijk | 27.0 | 0.20 | 1.4 | 5 | 2.40 | – | 0.152 |

**Table 2** Comparison of theoretical and numerical results for the extension test

| | $\sigma^0_{max}$ (kPa) | $\sigma_s$ (kPa) | $u_s$ (mm) | $u_c$ (mm) |
|---|---|---|---|---|
| Bilinear | 2400.00 | 600.00 | 0.0228 | 0.152 |
| Bilinear (numerical) | 2399.93 | 590.19 | 0.0249 | 0.153 |
| Hordijk | 2400.00 | – | – | 0.156 |
| Hordijk (numerical) | 2399.90 | – | – | 0.157 |

logged at one of the integration points in the cohesive element. Figure 13 shows the curves $w - \sigma_n$ and $u_p - \sigma_n$ for the extension test using the two studied softening curves. Initially, the contact presents a stiff behavior during the elastic regime and provides small to negligible deformations. After reaching the concrete tensile strength, the normal stress is dominated by the softening behavior. Both softening curves behave as expected showing smaller values for $\sigma_n$ as the contact plastifies. These results were compared with the theoretical values from both softening curves providing good agreement as shown in Table 2. Also it is worth to note that the overall behavior for both numerical results reproduces the theoretical curves qualitatively.

For the shear test, vertical displacements equal to $u = 0.187$ mm were applied to all nodes of the bulk element at right according to Fig. 14 aiming to cause failure due to shear stresses. The results are presented in Fig. 15. After analysis, we can conclude that the cohesive element model behaved as expected. Initially, the shear stress increases in an elastic regime up to a value close to the material shear strength. On the sequence, the contact plastifies reducing its capacity to transfer stresses between the bulk elements. Once again, according to Table 3, the values obtained from the numerical analyses were in good

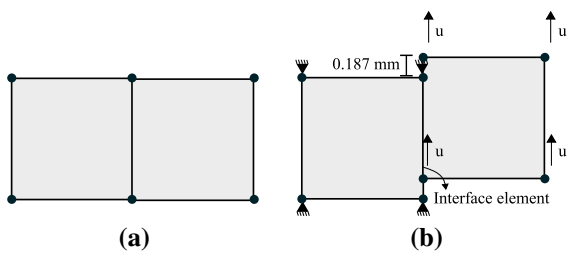
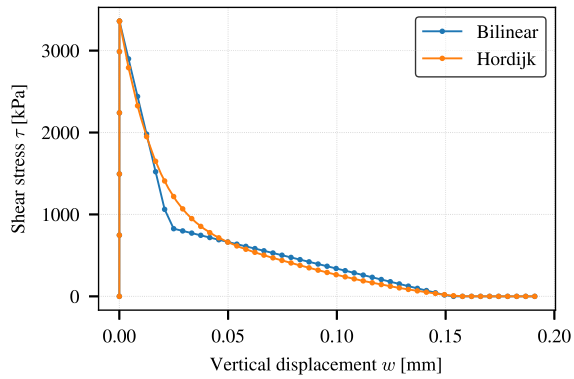
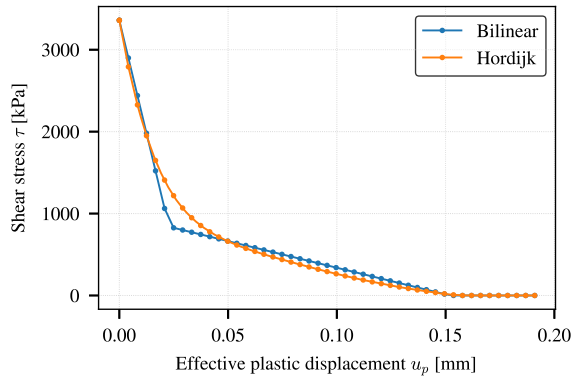



**Fig. 14** Shear test: **a** initial configuration; **b** deformed configuration




**(a)** Vertical displacement vs shear stress




**(b)** Effective plastic displacement vs shear stress

**Fig. 15** Shear test results

**Table 3** Comparison of theoretical and numerical results for the shear test

| | $\tau^0_{max}$ (kPa) | $\tau_s$ (kPa) | $u_s$ (mm) | $u_c$ (mm) |
|---|---|---|---|---|
| Bilinear | 3360.00 | 840.00 | 0.0228 | 0.152 |
| Bilinear (numerical) | 3359.26 | 826.07 | 0.0249 | 0.153 |
| Hordijk | 3360.00 | – | – | 0.156 |
| Hordijk (numerical) | 3359.00 | – | – | 0.157 |

agreement with the predictions from the softening curves. Also, it has been noted that the differences in the prediction of the maximum shear strength can be reduced by increasing the number of displacement increments.

### *4.1.2 Mesh objectivity test*

To verify that the application of the proposed formulation minimizes mesh dependency we present a simple mesh objectivity test similar to the one proposed by

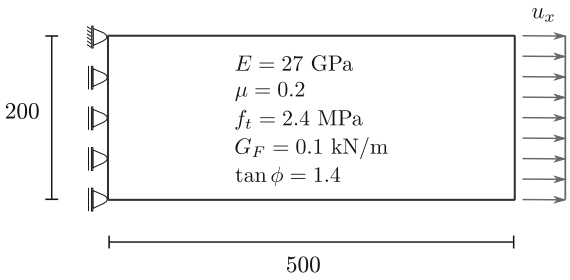



**Fig. 16** Boundary conditions for the mesh objectivity test

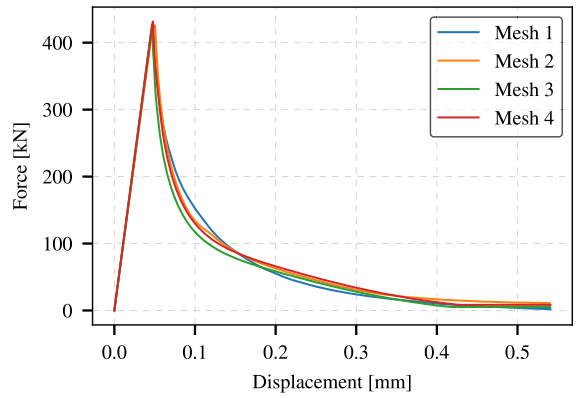


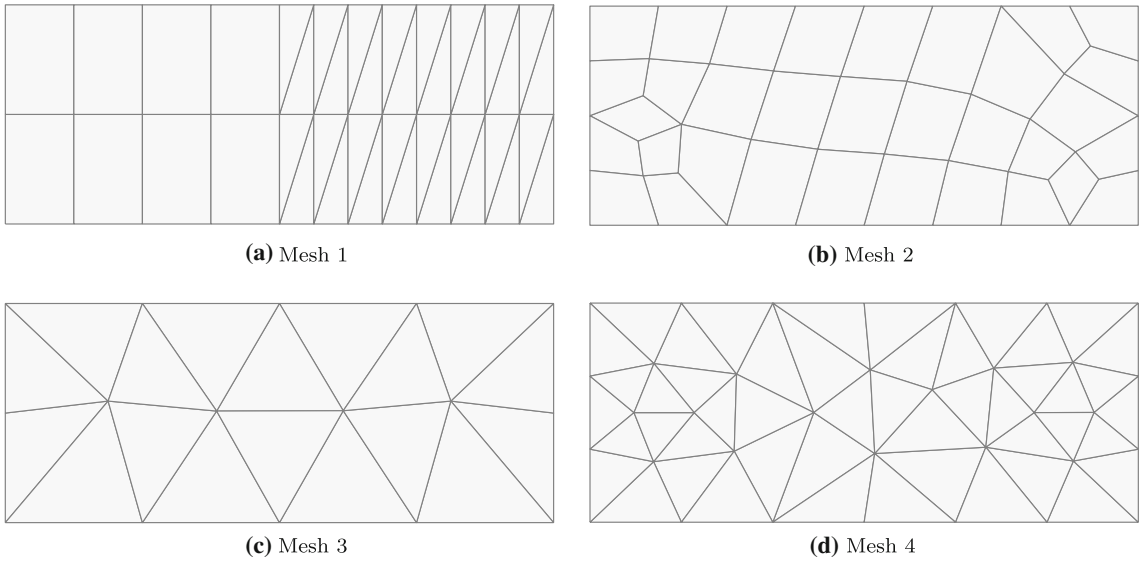

**Fig. 18** Numerical results using four different meshes

Donadon and Iannucci (2006). Figure 16 shows the domain, boundary conditions and material properties used in this analysis. Prescribed displacements $u_x$ at the right side were applied incrementally. Four relatively coarse meshes comprising triangular and quadrilateral elements were tested (Fig. 17). Although these discretizations, one structured and three unstructured, contain different numbers of cohesive elements we expect similar load–displacement curves before and after the peak load. Note that stiffness components from cohesive elements depend on the element's representative length in order to reduce mesh dependency; otherwise meshes with greater quantity of elements would present a more flexible behavior.

During the analyses, the prescribed displacement was logged as well as the resultant force. The load–displacement curves are presented in Fig. 18. As can be seen, before the peak load all curves follow pretty much the same path. These results demonstrate that the element representative length used to calculate the stiffness of cohesive elements significantly reduce mesh dependency on the elastic behavior. Also after the peak load and despite the use of coarse meshes, all curves are pretty close, showing the same nonlinear pattern.

## 4.2 Example 1—beam with a central notch

After validation, the proposed formulation is applied to the simulation of experimental tests. In this section, a beam with a central notch presented by Roesler et al.



**(a)** Mesh 1 **(b)** Mesh 2

**(c)** Mesh 3 **(d)** Mesh 4

**Fig. 17** Four different meshes studied in the objectivity test

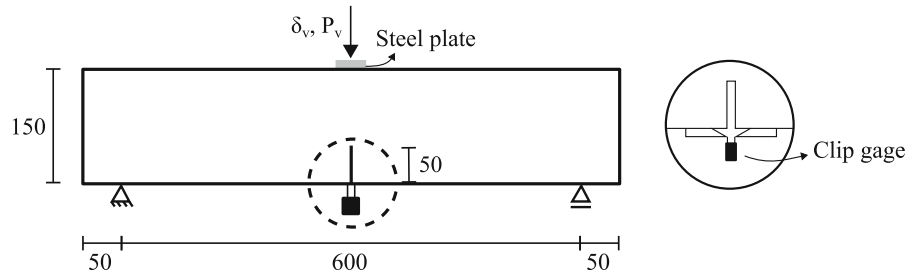



**Fig. 19** Test setup for a typical beam presented by Roesler et al. (2007). All measures are in millimeters

**Table 4** Material parameters for bulk and cohesive elements in Example 1

| Beam | $E$ (GPa) | $\nu$ | $\tan(\phi)$ | $\alpha$ | $\sigma^0_{max}$ (MPa) | $G_f$ (N/m) | $G_F$ (N/m) |
|---|---|---|---|---|---|---|---|
| V150-80 | 32.0 | 0.20 | 1.4 | 5 | 4.15 | 56.6 | 164 |

**Table 5** Mesh attributes for each model in Example 1

| Model | Bulk elements | Cohesive elements | Nodes | DOFs |
|---|---|---|---|---|
| B-T | 1808 | 2651 | 5432 | 10864 |
| H-T | 1808 | 2651 | 5432 | 10864 |
| B-Q | 2108 | 4108 | 8432 | 16864 |
| H-Q | 2108 | 4108 | 8432 | 16864 |

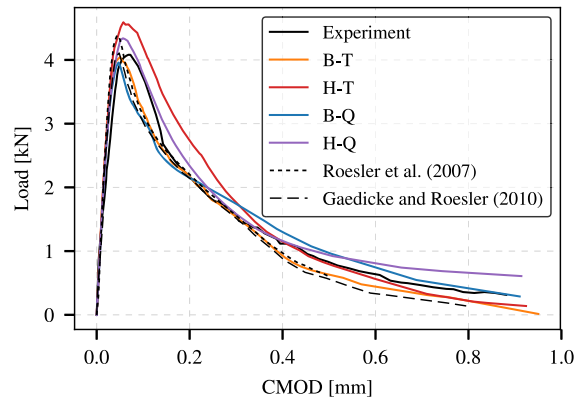



**Fig. 20** Experimental results and numerical predictions for Example 1

(2007) is studied. The test setup is shown in Fig. 19 where the specimen thickness is equal to 80 mm. The experimental test aimed to induce a single crack in mode I. For this purpose, a controlled displacement was applied over a steel plate located at the middle top of the beam.

For the numerical analysis, the bilinear and Hordijk's softening curves were used. The required material parameters are presented in Table 4. In order to promote some randomness in the shape of the crack path, an unstructured mesh was employed. Two types of elements were used to represent the concrete: quadrilateral elements with four integration points and triangular elements with a single integration point. For cohesive elements, four-node cohesive elements with two integration points were used. Table 5 presents a summary of all finite element models employed in this example. Letters B and H refer, respectively, to the bilinear and Hordijk's curves. Likewise, letters T and Q refer to triangular and quadrilateral elements, respectively.

During the finite element analysis, the reaction force due to the imposed displacement and the notch gap (CMOD) were logged for further analysis. Figure 20 shows the results of load versus CMOD for all models. Table 6 shows a comparative analysis of the peak load ($P_{max}$) and the corresponding CMOD ($\delta$) against the experimental results presented by Roesler et al. (2007). We can see that peak loads presented values close to the experimental data where most discrepancies were under 5% while the corresponding CMOD values presented greater discrepancies. Note that in this case, the use of the Hordijk's curve provided higher peak values. Also, the use of quadrilateral elements did not provide meaningful differences in the shape of the CMOD-load curve when compared to the results using triangular elements. In any case, the overall behavior of numerical curves was in good agreement with the experimental data.

In order to visualize the final deformation state, Fig. 21 shows the deformed mesh for model B-T at

**Table 6** Comparison of numerical and experimental results for Example 1

| Model | Increments | $P_{max}$ (kN) | CMOD $\delta$ (mm) | $\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ % | $\frac{\lvert\Delta\delta\rvert}{\delta}$ % |
|---|---|---|---|---|---|
| Experimental | – | 4.08 | 0.0683 | – | – |
| B-T | 1157 | 4.05 | 0.0506 | 0.8 | 25.9 |
| H-T | 1116 | 4.59 | 0.0574 | 12.5 | 15.9 |
| B-Q | 1006 | 3.96 | 0.0473 | 3.0 | 30.8 |
| H-Q | 1007 | 4.34 | 0.0565 | 6.4 | 17.2 |

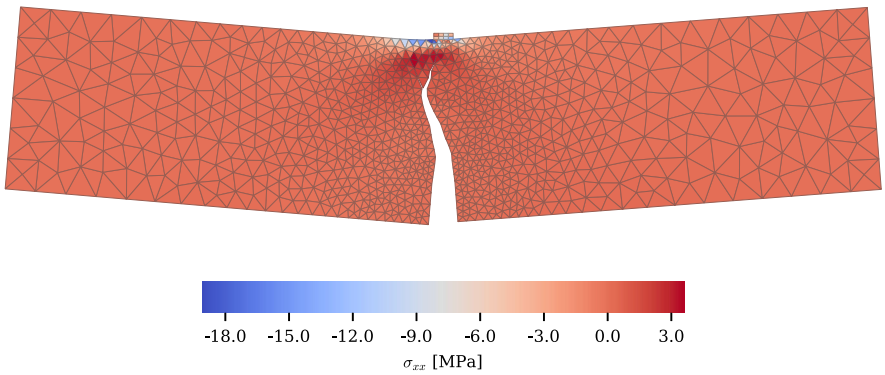



**Fig. 21** Crack path of simulation B-T of the beam in Example 1

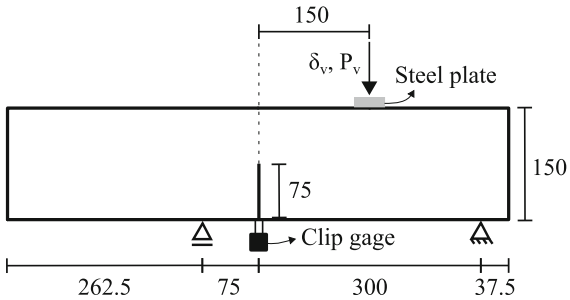



**Fig. 22** Notched beam with eccentric loading investigated by Galvez et al. (1998). All measures are in millimeters

**Table 7** Parameters of concrete and cohesive elements for Example 2

| $E$ (GPa) | $\nu$ | $\tan(\phi)$ | $\alpha$ | $\sigma^0_{max}$ (MPa) | $G_F$ (N/m) |
|---|---|---|---|---|---|
| 38.0 | 0.20 | 1.4 | 5 | 3.00 | 69 |

the end of the analysis using a scale factor of 50. Also, the $\sigma_{xx}$ stress field is displayed in the figure where it is possible to see a high concentration of tensile stresses at the crack tip, which is consistent with the fracture process.

Figure 20 also shows the numerical results from other authors using cohesive elements (Roesler et al. 2007; Gaedicke and Roesler 2010). Among the available results, the best ones from each author have been considered. In both cases, the curves show good agreement with the experimental data. Yet, it is worth to mention that the results from the first author were only available up to CMOD equal to 0.5 m.

### 4.3 Example 2—notched beam with eccentric loading

In this example, we simulate an experimental beam test investigated by Galvez et al. (1998). Figure 22 shows the test setup where the beam features a central notch and non-symmetric supports. The specimen's thickness is equal to 50 mm. This test aims to study the cracking properties of concrete in mixed mode; thus an eccentric displacement is imposed at the beam top with the aid of a steel plate. Also, in this case, the numerical analyses were performed using the bilinear and Hordijk's curves. The required material parameters are presented in Table 7. The values of Young's modulus, Poisson ratio, fracture energy and tensile strength of concrete were extracted from the work of Sancho et al. (2006).

For the finite element analysis, two unstructured meshes were generated. Those are composed of quadrilateral and triangular elements, respectively. Table 8 summarizes information for all studied models includ-

**Table 8** Summary of finite element models analyses in Example 2

| Model | Bulk elements | Cohesive elements | Nodes | DOFs |
|---|---|---|---|---|
| Mesh 1 | 264 | 383 | 792 | 1584 |
| Mesh 2 | 1056 | 1558 | 3168 | 6336 |
| Mesh 3 | 2376 | 3525 | 7128 | 14256 |
| Mesh 4 | 4224 | 6284 | 12672 | 25344 |

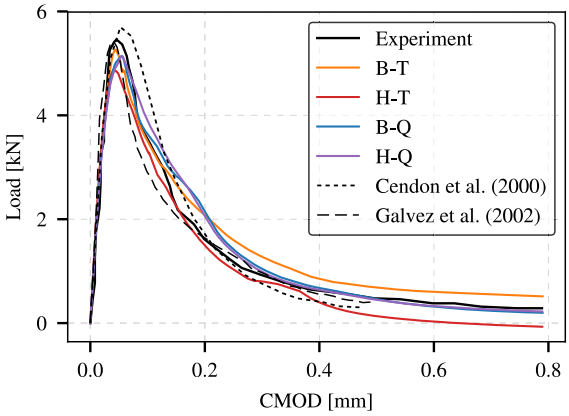



**Fig. 23** Experimental results and numerical prediction for Example 2

ing mesh data. Note that we chose to use unstructured meshes since they tend to provides crack patterns compatible with cracks observed in concrete specimens. Also, the authors noticed that the use of structured meshes, in some cases, provide increased strength and unrealistic crack patterns due to interlocking between bulk elements, especially in cases of mixed mode fracture. During the numerical analysis, the reaction force at the steel plate location, as well as the CMOD at the notch, were logged for further analysis. The load versus CMOD curves from numerical results are presented in Fig. 23 and compared with the experimental curves. From the results, we can observe excellent agreement between experimental and numerical curves, including at the softening branch. Note that all numerical curves follow the same pattern of the experimental one. Table 9 shows a comparison of the peak load ($P_{max}$) and corresponding CMOD ($\delta$) where, as in the previous example, it is possible to verify the very good agreement.

Figure 24 shows the beam at the end of the B-T analysis (model with the bilinear softening curve for cohesive elements and triangular elements for bulk ones) with a deformation scale of 50. The figure also shows the $\sigma_{xx}$ stress field where the stress concentrations are consistent with the loading and cracking process. Figure 23 also shows the numerical results from other authors (Cendón et al. 2000; Gálvez et al. 2002) that used cohesive elements. Again, the best results from each author have been considered. Although they were only available for CMOD values up to around 0.5 mm, both curves present good agreement with the experimental data and are similar to our results.

### 4.4 Example 3—double notched rectangular plate

The third experimental test to be analyzed was investigated by Shi et al. (2000). This test aims to study the fracture properties of concrete in Mode I by inducing cracks that start at each notch and go towards the center of the specimen. Figure 25 presents the test diagram, including the boundary conditions. As can be seen, the specimen is fixed at the bottom and features a steel plate at the top used to apply controlled displacements and, consequently, tensile stresses. The figure also shows the positions where displacements were logged with the aid of LVDTs. The specimen thickness was equal to 10 mm. Four finite element analyses were carried out as a result of combining triangular and quadrilateral meshes with the bilinear and Hordijk's curves. Table 10 shows the material parameters used in all analyses, and Table

**Table 9** Comparison of numerical results and experimental data for Example 2

| Model | Increments | $P_{max}$ (kN) | CMOD $\delta$ (mm) | $\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ % | $\frac{\lvert\Delta\delta\rvert}{\delta}$ % |
|---|---|---|---|---|---|
| Experiment | – | 5.46 | 0.0457 | – | – |
| B-T | 3396 | 5.27 | 0.0436 | 3.5 | 4.5 |
| H-T | 2586 | 4.89 | 0.0414 | 10.5 | 9.5 |
| B-Q | 1442 | 5.14 | 0.0540 | 5.9 | 18.1 |
| H-Q | 1389 | 5.14 | 0.0551 | 5.8 | 20.6 |

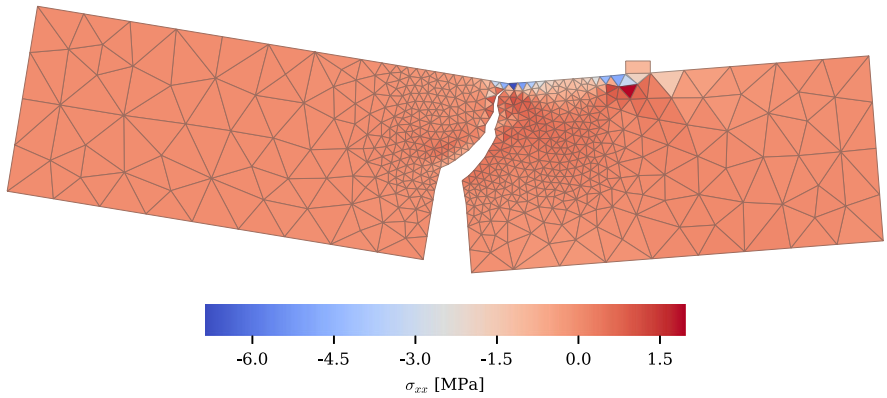



**Fig. 24** Crack path of simulation B-T in Example 2

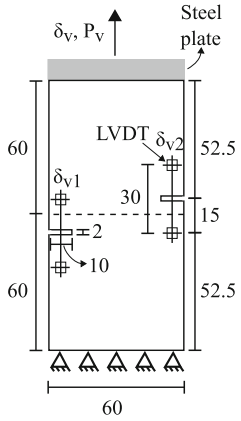



**Fig. 25** Test diagram for the double notched rectangular plate specimen. All measures are in millimeters

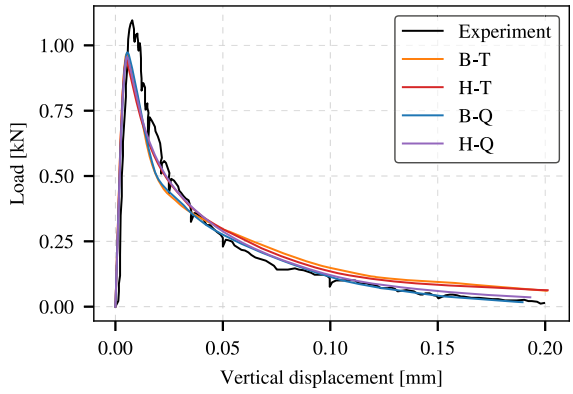



**Fig. 26** Experimental results and numerical prediction for Example 3

**Table 10** Material parameter for bulk and cohesive elements in Example 3

| $E$ (GPa) | $\nu$ | $\tan(\phi)$ | $\alpha$ | $\sigma^0_{max}$ (MPa) | $G_F$ (N/m) |
|---|---|---|---|---|---|
| 31.0 | 0.20 | 1.4 | 5 | 3.00 | 50 |

**Table 11** Summary of finite elements models analysed in Example 3

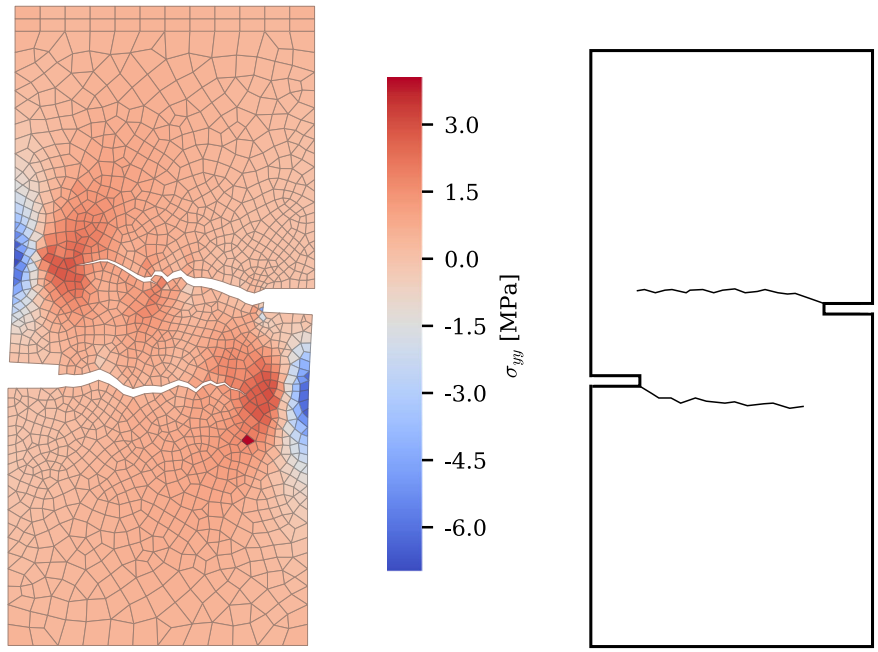
| Model | Bulk elements | Cohesive elements | Nodes | DOFs |
|---|---|---|---|---|
| B-T | 1830 | 2682 | 5490 | 10980 |
| H-T | 1830 | 2682 | 5490 | 10980 |
| B-Q | 1722 | 3181 | 6507 | 13014 |
| H-Q | 1722 | 3181 | 6507 | 13014 |

**Table 12** Comparison of numerical results and experimental data for Example 3

| Model | Increments | $P_{max}$ (kN) | $\bar{\delta}$ (mm) | $\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ % | $\frac{\lvert\Delta\bar{\delta}\rvert}{\bar{\delta}}$ % |
|---|---|---|---|---|---|
| Experiment | – | 1.10 | 0.0076 | – | – |
| B-T | 1712 | 0.97 | 0.0052 | 12.2 | 32.2 |
| H-T | 1392 | 0.95 | 0.0051 | 13.9 | 33.5 |
| B-Q | 1109 | 0.97 | 0.0056 | 11.5 | 26.8 |
| H-Q | 1132 | 0.96 | 0.0053 | 13.1 | 30.7 |

11 presents a summary of the studied finite element models.

During the finite element analysis, the reaction force $P_v$ at the steel plate was logged as well as the vertical displacements ($\delta_{v1}$ and $\delta_{v2}$) at positions consistent with the LVDTs locations, according to Fig. 25. Figure 26 shows a plot of the average vertical displacement, $\bar{\delta} = (\delta_{v1} + \delta_{v2})/2$, versus force $P_v$ while Table 12 presents a comparative analysis of the peak load and corresponding displacements from numerical and experimental results as in previous examples. Once again, very good agreement can be observed. Finally, Fig. 27 shows the deformed state multiplied




**(a)** Numerical analysis B-Q **(b)** Experiment

**Fig. 27** Crack path of double notched rectangular plate

by a scale factor of 50 at the end of the B-Q model analysis together with the experimental crack paths. Good agreement in the crack pattern is observed. Also, the figure shows the $\sigma_{yy}$ stress field where high tensile stress concentrations near the cracks tip are present.

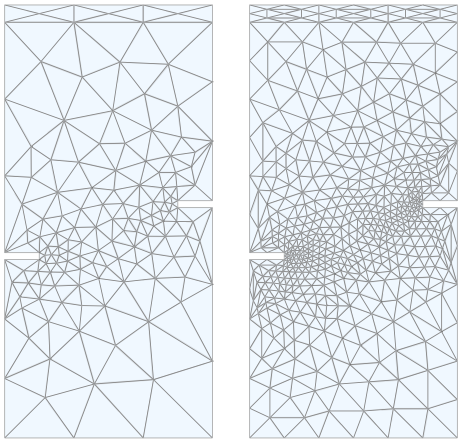
In order to verify convergence of numerical analyses using the proposed model, the same experiment is analyzed using different meshes. Initially, a coarse mesh is proposed. Later, three new meshes are obtained by refinement using the h-method. Thus, elements in these meshes are obtained by subdivisions of elements from the initial mesh. Figure 28 shows all four meshes, whereas Table 13 shows a summary.

The load–displacement curves for the four cases are presented in Fig. 29. As can be seen, curves follow a convergence pattern. Notice that results from Mesh 3 and Mesh 4 are quite close. This demonstrates that the application of the proposed model leads to convergent results.

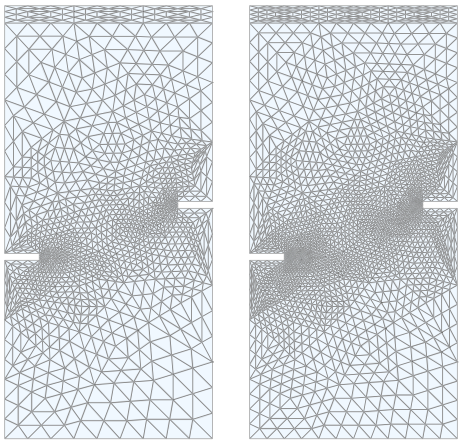
### 4.5 Example 4—double notched square plate

The fourth example analyzes one experimental test studied by Nooru-Mohamed (1992). Among several plain concrete specimens with different sizes and loading configurations studied by the author, we selected a 200 mm × 200 mm double notched square plate with 50 mm thickness where horizontal and vertical controlled displacements were applied simultaneously. The test induces slanted cracks that start at notches and go towards the center of the specimen with the purpose of studying the cracking properties of concrete in Mode I and II. Figure 30 shows the test setup. Note that the right side below the notch and the bottom of the specimen are bonded to a fixed steel plate. In turn, the left side above the notch and the top of the specimen are bonded to a movable steel plate. During the experimental test, a horizontal $\delta_h$ and a vertical $\delta_v$ controlled displacements (with equal rates) were simultaneously applied at the movable steel plate. Figure 30 also shows the positions where displacements were logged with the aid of LVDTs.

Four finite element analyses were carried out as the result of combining triangular and quadrilateral meshes with the bilinear and Hordijk's curves. Table 14 shows the material properties used in all analyses according to Nooru-Mohamed (1992) and Table 15 presents a summary containing the mesh attributes for all analyzed models. During the finite element analysis, displacements $\delta_v$ and $\delta_h$ were applied simultaneously at



**(a)** Mesh 1 **(b)** Mesh 2



**(c)** Mesh 3 **(d)** Mesh 4

**Fig. 28** Four different meshes studied in the h-refinement convergence test

**Table 13** Summary of finite elements models analysed in the h-refinement convergence test

| Model | Bulk elements | Cohesive elements | Nodes | DOFs |
|---|---|---|---|---|
| Mesh 1 | 264 | 383 | 792 | 1584 |
| Mesh 2 | 1056 | 1558 | 3168 | 6336 |
| Mesh 3 | 2376 | 3525 | 7128 | 14256 |
| Mesh 4 | 4224 | 6284 | 12672 | 25344 |

equal rates as in the experiment. The reaction forces $P_h$ and $P_v$ corresponding to the imposed displacements $\delta_h$ and $\delta_v$, respectively, were logged for further analysis. Also, the vertical displacements $\delta_{v1}$ and $\delta_{v2}$ and the horizontal displacement $\delta_{h1}$ were logged at positions compatible with the locations of LVDTs in the experiment (see Fig. 30). With this information, two plots were elaborated. Figure 31a shows a plot of the average vertical displacements, $\bar{\delta} = (\delta_{v1} + \delta_{v2})/2$, versus the vertical load $P_v$ and Fig. 31b shows a plot of the horizontal displacement $\delta_{h1}$ versus horizontal load $P_h$. As seen, both softening curves provide quite similar results. Also, most part of the numerical curves is in good agreement with the experimental ones. Table 16 presents a comparative analysis of the vertical peak load and corresponding displacements ($\bar{\delta}$) from numer-

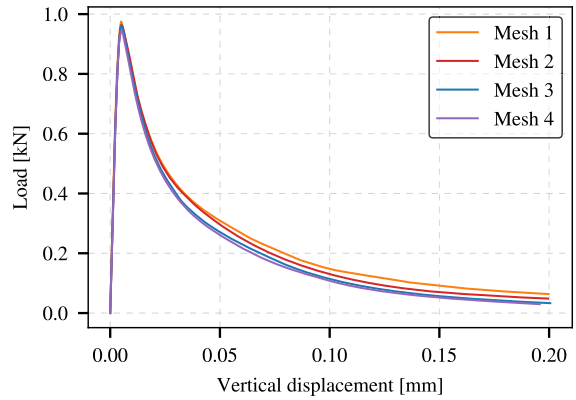



**Fig. 29** Results from the h-refinement convergence test using triangular elements

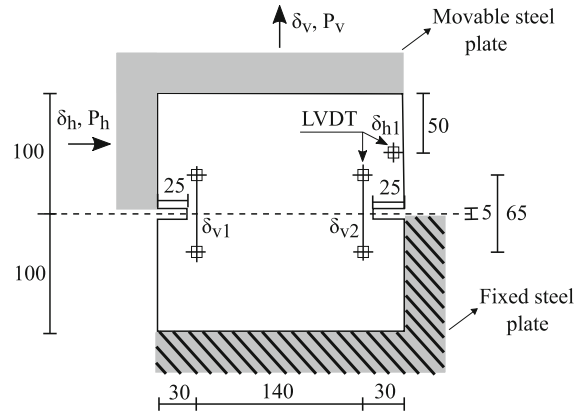



**Fig. 30** Test diagram for the double notched square plate. All measures are in millimeters

**Table 14** Material parameter for bulk and cohesive elements in Example 4

| $E$ (GPa) | $\nu$ | $\tan(\phi)$ | $\alpha$ | $\sigma^0_{max}$ (MPa) | $G_f$ (N/m) | $G_F$ (N/m) |
|---|---|---|---|---|---|---|
| 30.0 | 0.20 | 1.4 | 5 | 3.30 | 40 | 100 |

**Table 15** Summary of finite element models analysed in Example 4

| Model | Bulk elements | Cohesive elements | Nodes | DOFs |
|---|---|---|---|---|
| B-T | 3191 | 4739 | 9573 | 19146 |
| H-T | 3191 | 4739 | 9573 | 19146 |
| B-Q | 3281 | 6479 | 13124 | 26248 |
| H-Q | 3281 | 6479 | 13124 | 26248 |

**Table 16** Comparison of numerical results and experimental data for Example 4

| Model | Increments | $P_{max}$ (kN) | $\bar{\delta}$ (mm) | $\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ % | $\frac{\lvert\Delta\bar{\delta}\rvert}{\bar{\delta}}$ % |
|---|---|---|---|---|---|
| Experiment 1 | – | 16.80 | 0.0105 | – | – |
| B-T | 3534 | 16.92 | 0.0088 | 0.7 | 15.8 |
| H-T | 2016 | 17.01 | 0.0088 | 1.3 | 15.7 |
| B-Q | 4516 | 17.84 | 0.0105 | 6.2 | 0.5 |
| H-Q | 4008 | 17.88 | 0.0105 | 6.5 | 0.1 |

ical and experimental results. An excellent agreement is observed.

Figure 32a shows the deformed state with a scale factor of 50 at 75% of the H-T model analysis. In turn, Fig. 32b shows the experimental crack paths. Good agreement in crack patterns can be observed. Also, note that extra finite elements were required to represent the rigid steel plates where displacements are applied. Figure 32a also shows the $\tau_{xy}$ stress field where it is possible to see a concentration of shear stresses between the cracks due to the development of material discontinuity.

Figure 31 also shows the numerical results from Nguyen (2005) and Cusatis and Cedolin (2006). Although these authors do not use cohesive elements, their results provide a chance to compare the proposed model with other approaches. The first author uses conventional solid elements and simulates fracture by using a nonlocal damage model. In turn, the second authors use a confinement-shear lattice model where the material is represented by a set of connected particles. As can be seen in the figure, the method proposed by Nguyen (2005) was unable to predict the arise of negative vertical force along the softening branch. Also, the horizontal load curve was not well predicted. On the other hand, Cusatis and Cedolin provided better results, yet the curve for the horizontal load presents a pattern slightly different from the experiment.

### 4.6 Example 5—L-shaped panel

The last example analyzes the test investigated by Winkler (2011) where a 100 mm thick L-shaped specimen is loaded in order to study the concrete fracture properties in Mode I. Figure 33 shows the test setup. The bottom of the specimen is fixed, and a controlled vertical displacement is imposed in a small steel plate at the right side, aiming to induce tensile stress concentrations at point A.

For the finite element analyses, the material properties were taken from the work of Ghosh and Chaudhuri (2013) and are shown in Table 17. This test also aims to compare the performance of elements with linear and quadratic interpolations. A total of eight finite element analyses were performed by combining the two implemented softening curves with meshes composed by triangular and quadrilateral elements and different interpolation orders. Table 18 presents a summary containing the mesh attributes for all models. In this table LI, BI

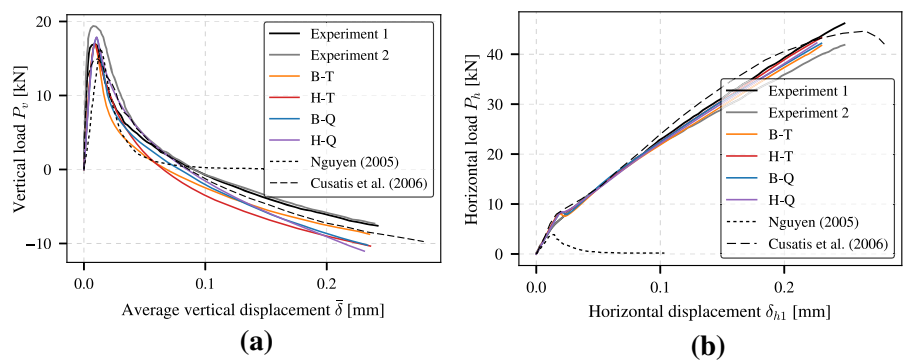
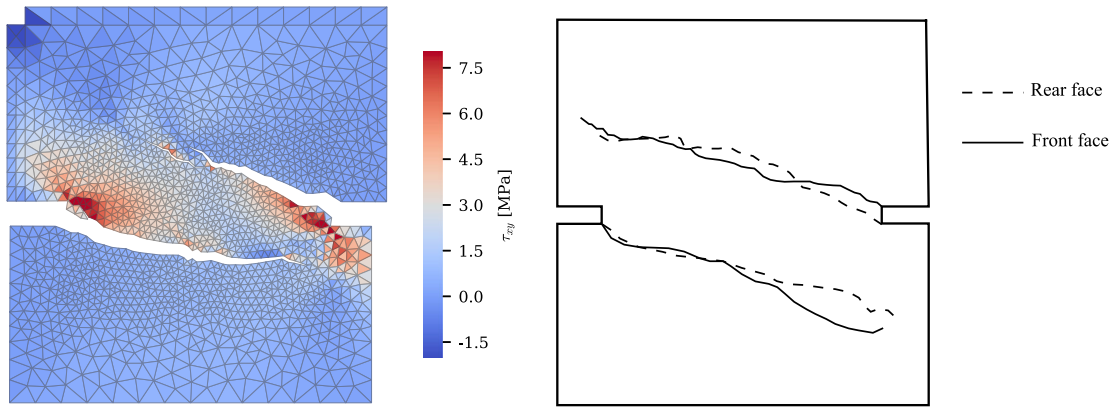



**Fig. 31** Experimental data and numerical prediction for Example 4




**(a)** Numerical analysis H-T **(b)** Experiment

**Fig. 32** Crack path for the double notched square plate

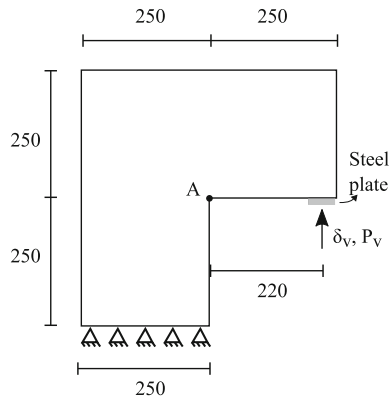



**Fig. 33** Test setup for the L-shaped panel specimen. All measures are in millimeters

and QI refer to linear, bilinear, and quadratic interpolations, respectively, used for the bulk elements. For linear interpolation, 3-node triangles with one integration point were used. In turn, for bilinear interpolation, we considered 4-node quadrilaterals with four integration points. Finally, for quadratic interpolation, 6-node triangles with three integration points and 9-node quadrilaterals with four integration points were employed. For the case of elements with linear and bilinear interpolation, the corresponding cohesive elements were linear with two nodes per side and two integration points. Likewise, for quadratic interpolation, the corresponding quadratic cohesive elements featured 3 nodes and 2 integration points.

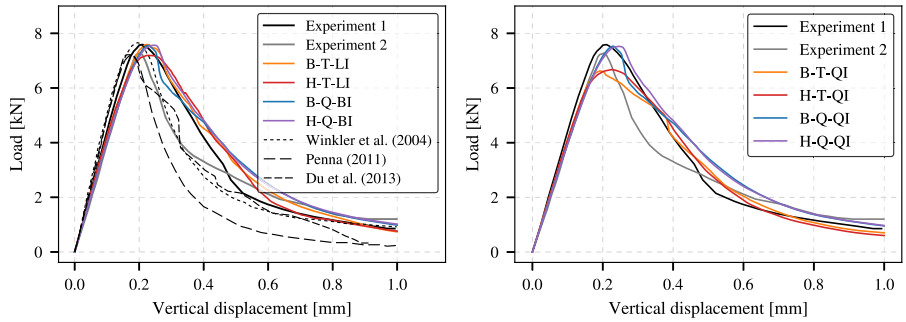
**Table 17** Material parameter for bulk and cohesive elements in Example 5

| $E$ (GPa) | $\nu$ | $\tan(\phi)$ | $\alpha$ | $\sigma^0_{max}$ (MPa) | $G_F$ (N/m) |
|---|---|---|---|---|---|
| 20.0 | 0.20 | 1.4 | 5 | 2.80 | 140 |

**Table 18** Summary of finite element models analysed in Example 5

| Model | Bulk elements | Cohesive elements | Nodes | DOFs |
|---|---|---|---|---|
| B-T-LI | 1673 | 2464 | 5021 | 10042 |
| H-T-LI | 1673 | 2464 | 5021 | 10042 |
| B-Q-BI | 3518 | 6956 | 14072 | 28144 |
| H-Q-BI | 3518 | 6956 | 14072 | 28144 |
| B-T-QI | 1673 | 2464 | 10044 | 20088 |
| H-T-QI | 1673 | 2464 | 10044 | 20088 |
| B-Q-QI | 3518 | 6956 | 31662 | 63324 |
| H-Q-QI | 3518 | 6956 | 31662 | 63324 |

In the course of the finite element analyses, the reaction force due to the imposed displacement was logged. Later, this force was plotted against the prescribed displacement and compared with the experimental data. Figure 34a shows the results with the use of linear and bilinear elements. Likewise, Fig. 34b shows the results plot for quadratic elements. From the figures, we con-




**(a)** Linear and bilinear elements **(b)** Quadratic elements

**Fig. 34** Experimental results and numerical prediction for Example 5

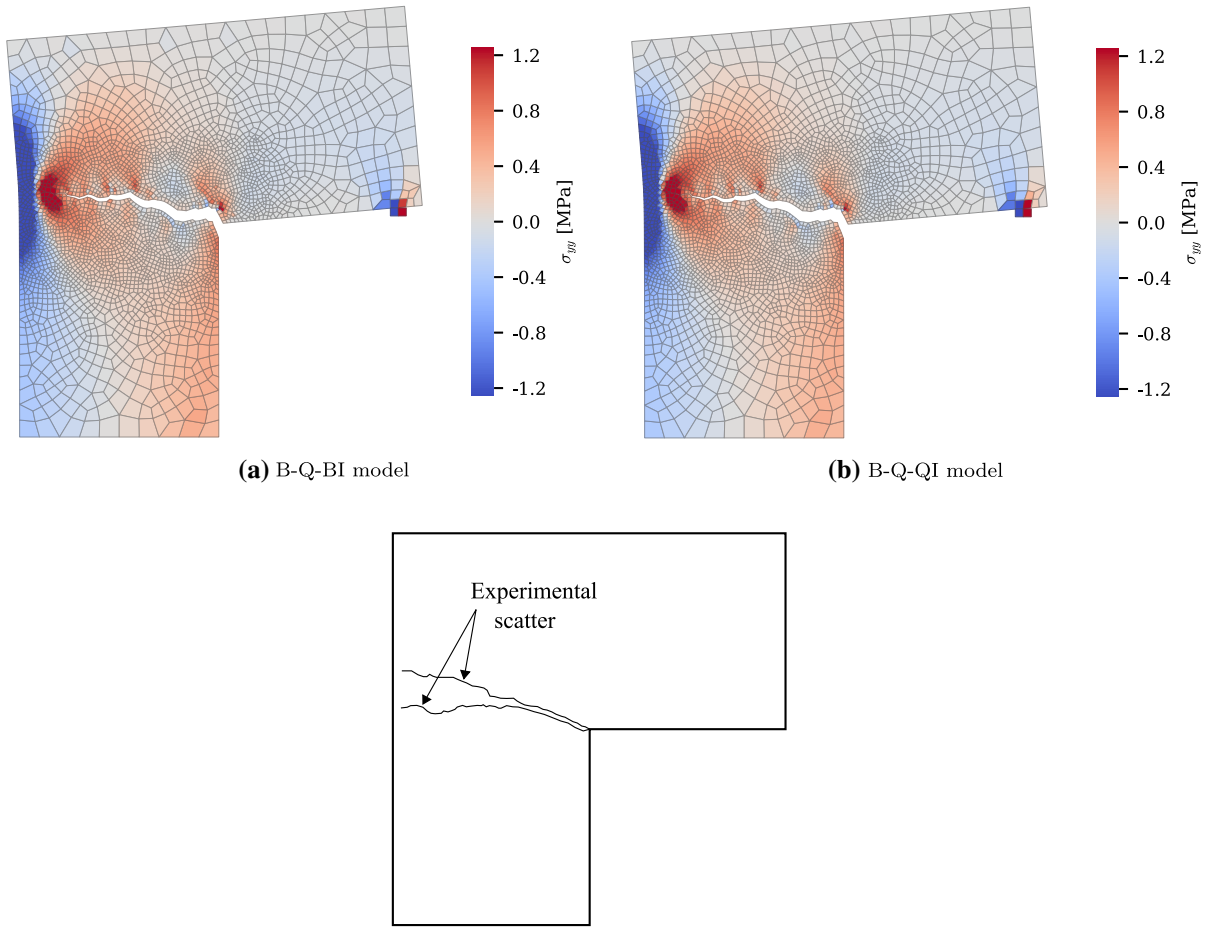
**Table 19** Comparison of numerical results and experimental data for Example 5

| Model | Increments | $P_{max}$ (kN) | $\delta_v$ (mm) | $\frac{\lvert\Delta P_{max}\rvert}{P_{max}}$ % | $\frac{\lvert\Delta\delta_v\rvert}{\delta_v}$ % |
|---|---|---|---|---|---|
| Experiment | – | 7.59 | 0.2112 | – | – |
| B-T-LI | 2205 | 7.60 | 0.2251 | 0.1 | 6.6 |
| H-T-LI | 2603 | 7.19 | 0.2244 | 5.2 | 6.2 |
| B-Q-BI | 1324 | 7.56 | 0.2260 | 0.4 | 7.0 |
| H-Q-BI | 1243 | 7.57 | 0.2463 | 0.3 | 16.6 |
| B-T-QI | 1581 | 6.63 | 0.1917 | 12.7 | 9.2 |
| H-T-QI | 1375 | 6.68 | 0.2218 | 12.1 | 5.0 |
| B-Q-QI | 1360 | 7.53 | 0.2290 | 0.8 | 8.4 |
| H-Q-QI | 1287 | 7.53 | 0.2427 | 0.8 | 14.9 |

clude that there is good agreement between the numerical results and experimental data. It is worth to note that there is no remarkable difference by the use of elements with linear and bilinear interpolation against their quadratic counterparts. Nonetheless, the use of quadratic elements presented a slightly more flexible behavior, especially during the loading stage, which is expected since quadratic elements present more deformation modes. Table 19 presents a comparative analysis for the peak load ($P_{max}$) and corresponding displacement ($\delta_v$) from all simulations against the experimental data. Again, despite the results from models B-T-QI and H-T-QI, the maximum load values show no significant differences between the use of linear and quadratic interpolations. Thus, we may conclude that, for this type of analysis, the quality of the results is more related to the discretization level rather than the order interpolation.

Finally, Fig. 35 shows a qualitative comparison of the crack paths for models B-Q-BI and B-Q-QI (using a scale factor of 50) against the experimental crack. Note that there is no significant difference between the crack paths due to the interpolation order. The figure also includes the corresponding $\sigma_{yy}$ stress fields where no significant differences are observed.

Regarding the comparison with other numerical results, Fig. 34a also shows the results from Winkler et al. (2004) that used a smeared crack model, Penna (2011) that used a damage mechanics model and Du et al. (2013) that used a formulation based on the XFEM method. When more than one result was available for each author, the best ones were considered in the figure. As can be seen, among these approaches, the XFEM method provided better results than the other despite presenting non-realistic oscillations in the softening branch of the load–displacement curve. On the other hand, our results seem to be closer to the experimental curves compared to the results of other numerical approaches.




**(a)** B-Q-BI model

**(b)** B-Q-QI model

**(c)** Experiment

**Fig. 35** Numerical and experimental results for Example 5

## 5 Conclusions

The paper presents a two-dimensional constitutive model for cohesive elements to be used in the simulation of cracking (mode I and mixed mode) in quasi-brittle materials. The model was elaborated using the plasticity theory featuring a Coulomb-based yield function, a non-associated flow rule, and an implicit stress integration scheme. Two softening laws were studied: a bi-linear law and the Hordijk's law. Finite element simulations using the proposed model work for fair levels of mesh refinement and elements with different interpolation orders. The model was validated in extension and shear tests and later applied in the simulation of several classical experiments of notched beams. In all cases, very good agreement between numerical and experimental results was observed in CMOD versus load curves, especially when observing the softening behavior. It represents an important outcome since the proposed method is able to simulate a wide range of experiments while it is reasonably simple to be implemented in computer codes. Regarding the aspects studied along with the examples, we can say that no significant differences in the results were observed by the use of different types of elements, although triangular elements provided slightly better crack patterns. Also, it has been observed that elements with quadratic interpolation did not significantly improve the results other than small changes in stress distributions. Regarding the softening curves, the use of a bilinear curve did not provide significant differences in terms of deformability nor crack patterns when compared to Hordijk's curve. Due

to the high nonlinearity of the fracture problem, the authors recommend the use of a finite element solver with automatic load/displacement stepping. Finally, the authors consider future research to extend the model to three-dimensional conditions.

**Acknowledgements** The support from the Brazilian Research Council (CNPq) is gratefully acknowledged.

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